

THE

English Moths and Butterflies

Together with

*The Plants, Flowers, and Fruits
whereon they Feed,
and are usually Found.*



BENJAMIN WILKES

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MOTHS AND BUTTERFLIES ***

Some minor changes to the text are noted at the [end of the book](#).

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Errata listed in front matter have been applied to the book.

Plates have been moved to be adjacent to the description of the species which they represent.

T H E
English Moths and Butterflies :

Together with
The Plants, Flowers, and Fruits whereon they Feed,
and are usually Found.

All *Drawn* and *Coloured* in such a Manner, as to *represent* their several
beautiful Appearances,

Being copied exactly from the Subjects themselves, and Painted on the best
Atlas Paper.

Together with an Attempt towards a *Natural History* of the said *Moths* and *Butterflies*.

THIS Work consists of one Hundred and Twenty Copper Plates, with a particular Account of the *Flies* represented in the said Plates, setting forth the true Times of their Appearance in the *Caterpillar*, *Chrysalis*, and *Fly* State; the most ready Means of finding them; the Method of managing and preserving them; their Ways of feeding; the several Plants they feed on; and, in a Word, every thing yet known relating to their *Natural History*. Together with the Names of the *Plants, Flowers, Fruits, &c.*

The Price of this Work colour'd is Nine Pounds;

Which for the Conveniency of the Buyer is divided into four Parts, and each Part, or any single Number, may be had separately.

Uncoloured Three Pounds Thirteen Shillings and Sixpence, or Two Shillings and Sixpence each Number.

The Natural History by itself Ten Shillings and Sixpence.

L O N D O N;

PRINTED for, and Sold by BENJAMIN WILKES, the Proprietor, in *Fleetstreet*.

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T O

MARTIN FOLKES, Esq;

PRESIDENT,

And to the

COUNCIL and FELLOWS of the

R O Y A L S O C I E T Y

O F

L O N D O N.

GENTLEMEN,

I Humbly take the Liberty to address this Work to You, and presume to hope, from Your exact Knowledge of the Subjects it contains, and Your having been pleased to countenance the several Parts of it, as they were published, with Your Acceptance and Approbation, that You will not, now they are put together, deny them the Honour of Your Patronage.

The Institution of Your Society for the Improvement of Natural Knowledge, must render it not improper to lay before You every Attempt

towards such Improvement. On the contrary, You have a sort of Right to expect, that every Author in Natural History, should submit his Labours to Your Examination, and think himself happy if they will stand the Test of Your Judgment.

The Animal and Vegetable Parts of the Creation afford endless Stores for the Enquiry and Entertainment of the curious Naturalist: and some of the most beautiful of each Sort, are here presented to the Eye, in their true Size, Form, and Colour. Nor will it be objected by You, that too much Time and Pains has been bestowed on a Company of Insects, which few People trouble themselves about; for true Philosophy teaches, that the meanest thing God has made, is not below the Dignity of the most exalted of mortal Men to consider. But the Creatures here exhibited, are adorned with such a Variety of Beauty to engage our Notice, and undergo such amazing Changes in their Form and Appearance, that a thinking Mind can hardly avoid regarding them with uncommon Pleasure and a more than ordinary Attention.

Little, perhaps, is wanting to make the Study of Nature universally pursued, than to render it easy and intelligible; for, surely, the Understanding God has given us, can never be more properly employed, than in examining and considering those Proofs of his Divine Goodness and Power, which are placed every where around us. This is a most reasonable and satisfactory Entertainment; and I am fully persuaded, many of You, Gentlemen, have often experienced more real Pleasure from such Examinations, than those who know nothing of them can possibly conceive.

My Attempt towards a Natural History of the *English Moths* and *Butterflies*, (which is here added to the Paintings of them) wherein they are divided into Classes, and their several Progressions and Changes from the Egg to the Fly-state are described, (and that not from Books or Conjectures, but from actual Experiments and close Observation) will, I hope, meet with Your Approbation.

But not to be longer troublesome than to beg Pardon for this Presumption, and with a continual increase of Prosperity and Honour to the ROYAL SOCIETY of *London*, permit me to subscribe myself,

GENTLEMEN,

Your most obedient,

Humble Servant,

BENJAMIN WILKES.



T H E
P R E F A C E.

THE Painting of HISTORY PIECES and PORTRAITS in Oyl being the Profession of the Author of this Work, he frequently found himself at a Loss, in the Course of his Employment, to understand with Certainty what Colours would contrast with and set off each other to the best Advantage; and perhaps had still remained under the same Difficulties, had he not been led by Accident into the Means of learning all this from Nature, which undoubtedly is the best Instructor.

IT happened that a Friend invited him one Evening, to bear him Company to a Society named the AURELIAN: the agreeable Purpose of which is, to become acquainted with that Part of Natural Philosophy which relates to the Generation and Properties of Insects, of which they have a fine Collection. And here he first saw such Specimens of Nature's admirable Skill in the Disposition, Arrangement, and contrasting of Colours (particularly amongst the Moths and Butterflies) as struck him with Amazement, and convinced him, at the same Time, that studying them would turn greatly to his Advantage.

WITH this View he desired to become a Member of the AURELIAN SOCIETY, and soon after had the Honour to be admitted; when, by his own Diligence, together with the genenerous and kind Assistance of some Gentlemen of the Society, it was not long before he had acquired a tolerable Collection of English MOTHS and BUTTERFLIES; and for ten Years past his leisure Hours have chiefly been employed in the collecting and making Drawings of the different English Caterpillars, Aureliæ or Chrysalides,

Flies, &c. that he could any Ways obtain; and in making such Observations and Remarks thereon as shall now, with all Submission, be laid before the Publick.

H E has been greatly assisted in this Work by that well-known and ingenious Naturalist, Mr. Joseph Dandridge; to whose noble Collection he had free Access during his Life-time, and also the Liberty of making what Use he thought fit of his curious Remarks on these Subjects, which were the Fruits of no less than forty Years Experience. And he is still favoured in the like Manner by his Successors, as often as he finds Occasion.

R E S P E C T to the Memory of this valuable and worthy Man, will not suffer him to omit this Opportunity of giving him the Character that is most justly his due, viz. That he was a Person of unbounded Curiosity, and unwearied Application in his Researches into the Works of Nature; in the Study of which all those Hours were employed that could be spared from his immediate Business; by which Means, during a long Life, he had not only got together a great Variety of the most rare natural Productions, but also from Time to Time had written down his Observations thereon, which deserve the highest Regard, as his Judgment and Veracity were both without Exception. And, as he loved these Studies himself, he was a zealous Encourager of all that had the same Inclination, whom he was always ready to assist with his laboriously acquired Knowledge.

B U T to proceed—He must also acknowledge, that in the Course of this Undertaking he has made use of the Drawings of some Caterpillars and Flies which were published not long ago in a curious Work of this Sort, by J. A. Rosel, at Norimburgh, in Germany: But he has taken from him such only as are exactly of the same Species with what we have in England, though we have not yet been so fortunate to meet with them under all their several Changes: what Information that Gentleman has given, as to those Particularities concerning them which we are yet unacquainted with, he therefore judged proper to insert.

A S to the Plants, Flowers, and Fruits, he declares himself under the highest Obligation to a most curious Naturalist, and worthy Member of the Royal Society of London, whose costly and valuable Collection is known to the learned World, and esteemed as it deserves. This Gentleman has generously permitted him to make use of many excellent Paintings of these

Subjects, which were taken from Nature by those two celebrated Artists Mr. George Dennis Ehret, and Mr. Jacobus Van Heysum, and executed with the utmost Judgment and Command of Pencil, so that he could hardly doubt but that true Copies of them would give great Pleasure to the Lovers of Botany, whom he has endeavoured to entertain with all the Variety he could introduce.

SOME Ill-natured People, that love to find Fault with every Thing, have, he hears, objected against the Liberty he has taken, in giving in several of his Plates, Plants, Flowers, and Fruits, that are not in Reality the Food of the Caterpillars therein exhibited; but as the greatest Part of the Caterpillars described in this Work feed chiefly on the Oak, Elm, Black-Thorn, White-Thorn, Willow, and Nettle, all which are separately represented in different Plates, it was judged unnecessary (and would certainly have been tiresom) to repeat those Subjects, as often as different Caterpillars are presented to which they serve for Food: and therefore, instead of so doing, the Author has enriched his Undertaking with some of the most beautiful Productions of Nature in the Vegetable Kingdom: But, however, has not forgot to place the Caterpillars, in most of the Plates, on a small Specimen of the real Plant on which it feeds, as will be properly explained in the Account of each Particular Fly:—and this Care, it is hoped will be an Answer to all unreasonable Cavils.

THE Plates are all drawn and etched with his own Hand, in a quite new Manner; whereby every Design, when coloured, appears like a regular Piece of Painting; and the Flies are drawn and Painted with the utmost Exactness, from as fine and perfect Specimens as can be bred. The common Method of amusing the Reader with long Accounts of the Colours and Markings of each Fly is avoided in this Work, as entirely needless, when the Objects are truly represented before his Eyes.

HIS Classing of the Caterpillars will be found of great Use; for if it be enquired what Fly any Caterpillar that may happen to present itself will produce, by observing its Character you will easily be satisfied (if it is in this Collection) and be likewise taught what its proper Food is, &c. Names are absolutely necessary to distinguish one Fly from another, and enable the Curious in this Study to communicate any Observations they may make; which, without giving Names to the Flies, would be very difficult, if even possible to be done; as they must be greatly at a Loss to make others

perfectly understand what Flies they are treating of. The worthy Members of the AURELIAN SOCIETY, and several other ingenious Gentlemen, have been so sensible of this, that they, from Time to Time, have thought proper to bestow some Name on every Species that has come within their Knowledge.

I N D E E D Names are arbitrary, and if remembered, and the Intent of them well understood, may, whatever they are, serve the Purpose of giving them: But here, as in every Thing else, a Propriety is to be aimed at, and the Names given ought to denote some distinguishing Particularity, and that the most obvious, in the Subject on which they are bestowed, so as to make the Name, if possible, descriptive of the Thing: Which Rule has been followed as strictly as might be in naming the Moths and Butterflies contained in this Work: that is to say, the Shape, the Colour, the Marking, the Food, or the Place where found is what they are named from. But in doing this, where the Author found them named to his Hands in this descriptive Manner, he has gladly retained the Name they had already, and only presumes to bestow Names himself on such as either had none at all, or which he thought not sufficiently denotative of the particular Flies intended to be known by them.

T H E Division of the Work into Books and Chapters may need some Explanation, since under some of the Divisions very few, or even but one single Fly is placed; the Reason whereof is, that the Plan here laid down is intended for the Arranging of all the Moths and Butterflies, whose Progression is known at present, or shall hereafter be discovered in England; of which, as the Flies here given are but a small Part, this must be considered but as an Out-line to be filled up occasionally.

I T is hoped the Directions here given how to collect the Caterpillars or Chrysalides, how to feed and take Care of them, how to preserve the Flies when bred, and where and at what Times they may be found most easily, will prove both acceptable and useful to every curious Collector; and as the Author thinks himself under great Obligations to the Subscribers to, and Encouragers of this Undertaking, a List of their Names is added as a Memorial of his Gratitude.

T H E S E Things being premised, we come now to give a short Sketch of the Natural History of the English Moths and Butterflies in general.



I N T R O D U C T I O N.

Of MOTHS and BUTTERFLIES in general.

THE Opinion of *Equivocal Generation* however formerly received, has by the present Age been deservedly rejected; since every Observation proves, that all Creatures are produced by Parents of their own Kind, and that in general those Parents are Male and Female; and the Production of every *Moth* and *Butterfly* in this Manner, is I believe at present not doubted by any Body.

THE Females, both of the *Moth* and *Butterfly*, lay their Eggs in a short Time after Copulation, *upon* or contiguous *to* what will supply proper Food for the young Brood when produced. After a certain Period the Infant *Caterpillars* included in the Eggs burst their Shells, crawl forth, and feed upon the Provision ready prepared for them. And each of these *Caterpillars*, having fed its appointed Time, becomes a *Chrysalis* or *Aurelia*; from which State, after a while, it issues forth a *Fly* in the Likeness of its Parent.

SOME Species of *Butterflies* breed twice a Year. Those that come forth in *May* lay their Eggs soon after, which Eggs in about nine Days Time are hatched. The *Caterpillars* feed for about six Weeks, then change to *Chrysalis*, lie in that State for about fourteen Days, and then the *Flies* are bred. These lay their Eggs as the first Brood did, the *Caterpillars* change to *Chrysalis* in *September*, and the *Flies* come forth in the following *May*.

THE Variety of the *Moth* Kinds is so great, that perhaps there is scarce a Day in the Year when some of them are not bred. The Time of their Continuance in the *Chrysalis* State will be best known by attending to the Account of each *Moth* described in the following Plates. Some *Moths* as well as *Butterflies* breed twice a Year.

IT frequently happens, that the *Fly* does not come out at the usual Time, but continues in the *Chrysalis* State till that Time Twelvemonth; so that the Creature remains in that Case a Year and nine Months (and sometimes longer) in a State of Rest, without Nourishment of any kind.

THUS much in general:—We descend now to particular Observations.

On the different Manner of laying their Eggs.

A L L *Butterflies* and *Moths* lay a great Number of Eggs, some even as far as two or three Hundred. The Eggs of some Kinds are glutenous, or covered with a glewy Moisture; but those of others are not so: The glutenous are fixt (sometimes singly) on *Grass, the Leaves of Trees, Shrubs, &c.* and sometimes in Clusters after the same Manner. Some Sorts of Eggs are laid naked or uncovered, and others are cloath'd with a sort of Down, &c. Some *Flies* lay them in a Spiral Line, or Screw-like Figure, round a small Twigg, a Stalk of Grass, &c. And these may be pulled off entire, in Fashion like a Cylindrical Tube, and as hard as Horn: Others deposit theirs in the Chinks of the *Barks of Trees, &c. &c. &c.*

IT is observeable, that the whole Process from the Egg, to the *Fly*, of many Species, is perform'd in about sixty Days. But the *Goat Moth* is an Exception to this Rule, being reported to be three Years proceeding from the Egg to the *Fly* State.

Of the Eggs.

T H E Eggs are of various Forms; some are round, some are oval, and many of the *Moth-Kind* flat, and perforated through the Middle; so that the *Embrios* or *Caterpillars* lie curl'd up before they are hatch'd, as they frequently do afterwards when come to Maturity. They are found of all manner of Colours, and many of them are extremely beautiful if view'd with a Microscope; some being curiously rib'd and adorn'd with *Protuberances, &c.* like the nicest carved Work; others marbled and spotted with an amazing Variety of delightful colouring.

Of CATERPILLARS.

CATERPILLARS are usually divided into the Naked and Cloathed Kinds. Among the Cloath'd some have only small Tufts of Hair or Down, others are cover'd all over with it.

THE *Caterpillars* of all known *Butterflies* have six Hooks or Claws before, eight Feet in the Middle of their Body, and two Holders behind. Those that produce *Moths* have all six Hooks or Claws before, and Holders behind. Some have eight Feet in the Middle, some four, and some only two next their Holders.

SOME Species of *Caterpillars* always feed inclos'd or spun up in the Leaves they feed upon, others take their Food openly and in Sight.

SOME are sociable and herd together whilst young, but when they change their third Skin they grow unsociable and feed separately.

OTHERS are born unsociable, and feed separately from their Infancy.

SOME Species spin a large Web that will contain an hundred *Caterpillars*, or more, from which they never go far till they have eat up all the Leaves that are near thereto: Then they spin a new Web, and so (shifting their Quarters) in a short Time, instead of the green Leaves that they have devoured, leave the Tree, or Bush, covered only with their white Webs.

SOME Sorts, when disturbed, let themselves down by a Thread like a Spider, by which Means they avoid being devoured by other Insects that have not the Means of following them, &c.

Of the Food of CATERPILLARS, and their Manner of Feeding.

THEIR Food is almost general, but some Kinds will eat nothing except their own particular Trees, or Plants.

SOME Species feed upon Herbage, others live and feed upon the solid Parts of Trees, and Barks.

SOME dwell in the Earth, or other private Recesses in the Day-time, and at Night come forth, and feed on Grass, Flowers, &c. Others again feed on the different Kinds of Mosses, Heath, Broom, the Leaves of Trees, Shrubs, &c.

SOME feed upon the Waters naked and exposed, and others make themselves Cases of Sticks, Rushes, &c. in the Waters, where they get their Food.

Of the Forms of CATERPILLARS.

THE Variety and Beauty which present themselves to our Eyes in the Form and Colouring of *Caterpillars*, are no less remarkable than the *Flies* themselves, as the Reader will be satisfied by examining the following Plates.

Of the Change of CATERPILLARS into the *Chrysalis* State.

WHEN a *Caterpillar* has fed its appointed Time, and is come to full Maturity, it undergoes a Change, and becomes a Body of a different Form, incrustured with a Shell, and usually called a *Chrysalis* or *Aurelia*; the Shell of which is formed underneath the Skin of the *Caterpillar*, which Skin comes off as soon as the *Aurelia* becomes perfect.

Of the CHRYSALIS or AURELIA.

DURING its Continuance in the *Chrysalis* State, it seems to have no Sense but Feeling; and some of them have even that in a very low Degree, for they may be touched pretty roughly without moving.

THE *Chrysalides* or *Aureliæ* of the *Butterfly-Kind* are all very beautiful; some of them look as if they were studded with Gold; some are striped, others spotted, and others mottled with all the various Colours of the Rainbow. But those of the *Moth-Kind* are not so; differing very little from

one another, either in Shape or Colour, except the Magpye, and some few besides.

How the *Aurelia* or *Chrysalis* is secured.

NOTHING in Nature is perhaps more deserving Notice, than the many different Methods whereby *Caterpillars* conceal and secure themselves whilst in this helpless and unactive Condition.

SOME of them spin a large Case of Silk in Form of an Egg, wherein the *Chrysalis* lies concealed, till the Time comes for the *Fly* to make its Appearance, which it does by piercing through this Case.

OF such Cases there are several Sorts, different in Form and Colour:

SOME are long, and drawn out to a Point at each End; others are more round.

THE Texture of some is very thick and close.

OTHERS are thin, like Gause, insomuch that the *Chrysalis* may be seen thorough them.

SOME *Aureliæ* are white, some yellow, and others of different Hues; and most of the hairy *Caterpillars* intermix their Hair amongst the Spinning.

SOME *Caterpillars* form a hard-crusted Case, composed of silk Threads glewed together, in such Manner as to become hard like an Egg-shell, so that when the *Fly* comes out, a round Hole like a Door appears.

THE Silk-spinning *Caterpillars* are of various Kinds, and deposit their *Aureliæ* in different Places, and in different Manners, viz. some fasten them on the Leaves and Stalks of Plants, without any Covering; others spin the Leaves over them; others lodge them in Holes and Crevices; and others in the Earth.

MANY of the *Moth-Kind* place their *Chrysalis* in the Earth, about an Inch or two deep, surrounding it with a Case of Earth lined with a glutinous Matter, that no Wet can penetrate: And what is very surprizing, notwithstanding the extreme Delicacy and Tenderness of the *Fly* when first

bred, be the Ground never so hard, it makes its way thorough it without ruffling a single Feather.

FEW, if any, of the *Butterfly-Kind* form a Case or Spinning, except what fastens the Tail and Body of the *Chrysalis*. Some hang perpendicularly downwards, and others in an horizontal Position, being fastened by the Tail, with a Thread round the Back and on each Side. Some are secured within the Leaves spun together; others are hid in the Hollows and Barks of Trees; and others again under the Copings of Walls, in Houses, Sheds, &c.

Of the F L Y .

AFTER the *Chrysalis* has remained its due Time, the *Fly* bursts the Shell wherein it was contained, and comes out formed perfectly in every Part, except the Wings, which then appear too small to cover its Body. In this naked Condition it crawls up to some convenient Place, where its Wings may hang down without any Hindrance: And when thus disposed, the Wings begin to expand themselves so much, that in the Space of half an Hour, the Wings (of *Butterflies*) are stretched commonly to their full Size; nay, even those of the largest *Moths* seldom require above an Hour's Time. At first they are flabby, like wet Paper, but soon afterwards they become stiff, and capable of supporting them in their Airy Flights, &c.

IN this *Fly* State their first Care seems to be the Propagation of their Species, for some of them will copulate immediately if they can find a Mate; to seek which the Male *Butterfly* takes Wing, as soon as his Wings are in a proper Condition: But few *Moths*, whether bred in the Night Time or in the Day, ever offer to fly till the Night after their Birth, when they go in Search of the Females.

THE Females (especially of the *Moth* Kinds) seldom, if ever, take Wing till they have been cocked: they often continue in Conjunction twelve Hours and more, after which the Female lays her Eggs, flies about for a Time, and then dies.—The Females of some Kinds are without Wings.

THE Males (of some Species) have a surprizing Quickness and Distinction in their Sense of Smelling, exceeding that of the Blood-Hound, or any other Creature yet known; for if a Female be concealed in a Box, and

there are any Males within half a Mile or more, they will fly directly to the Place, and hover round it. By this Means a Dozen Male *Egger-Moths* have been taken in an Hour's Time, and that at about Twelve o'Clock at Noon, as may be seen in the Account of the *Egger-Moth*.

THEIR Continuance in the *Fly* State depends greatly on the Weather, many *Flies* being destroyed as soon as bred, by Storms of Hail, &c. but if the Season proves favourable, many will live for a Fortnight, three Weeks, or longer, and some even all the Winter through.

The Distinction of MOTHS and BUTTERFLIES.

MOTHS and BUTTERFLIES agree in their Progression from the *Egg* to the *Fly* State, are different when in that State, in the following Particulars.

BUTTERFLIES have a Ball or Knob at the Extremity of each of their *Antennæ*, or *Horns*.

THE Horns of MOTHS have not such Balls or Knobs, but are constantly threaded or pointed: except the Males of some Species, whose Horns are ramified, or in the Form of Combs.

ALL BUTTERFLIES settle with their Wings erect, and generally keep them in that Posture, unless when basking in the Sun.

MOST MOTHS do not settle with their Wings erect, but place them horizontally, or inclining to their Feet.

ALL BUTTERFLIES come abroad by Day-Light, and chiefly when the Sun shines bright.

MOTHS fly, some by Day-Light, some by Twilight, and some in the Dead of Night.

N. B. THERE is a Species of *Fly* betwixt the *Moth* and the *Butterfly*, whose Horns are more flat and hollow; this is describ'd in the second Book of this Work, by the Name of the *Burnet*.

Concerning the Food of the FLY.

THE Food of these Creatures in the *Fly* State, is very different from what it was when they were *Caterpillars*: *Butterflies* feed on all Kinds of Flowers, and on several sorts of Fruits, and I have seen them extremely fond of the Juices that issue from the Bodies of several Sorts of Trees.

MOTHS feed on the same Things, as also on the Honey-Dew that is found on the Leaves of Limes, Elms, &c. The *Moths*, as well as the *Butterflies*, are furnished with a long Proboscis, or Tongue, which lies curl'd up like a Watch-Spring, under the fore Part of the Head. The Length of this Proboscis varies according to the Species of the *Fly*, and is from half an Inch, to four Inches long.

The Enemies of *Moths* and *Butterflies*.

NATURE has been so abundant in the Production of these Creatures, that were it not for the many Dangers they are exposed to, through every Stage of their Lives, the whole World would be too small to contain their Offspring, in four or five Years; for, upon Calculation, the Progeny of one Pair of large *Butterflies*, (supposing them to lay only 100 Eggs, whereas many of them lay 2 or 300, and that twice a Year) would be sufficient to cover the whole Kingdom of *England* in about twelve Years.

WHILE in the Egg, they are devoured in great Numbers by Birds and other Animals.

WHILST *Caterpillars*, they are a Prey to small Birds of all Kinds, being almost the only Food wherewith such Birds nourish and rear their young ones.

CATERPILLARS likewise kill and devour one another; and, besides that, Numbers of them die in shifting their Skins, and in changing into the *Chrysalis* State. But the greatest Enemy they have in this State, is the *Ichneumon Fly*.

IN the *Aurelia* State, such as have no Case are a Prey to Birds, &c. those in the Earth, to Moles, Mice, and other subterraneous Animals; and besides, great Numbers are destroyed by Frosts and Wet. In the *Fly* State they are a Prey to Birds, Batts, and many other Creatures.

THERE are four different Ways, in one or other of which all Animals of the *Moth* or *Butterfly-Kind* subsist during the Winter Season, viz.

1. SOME in the Egg State.
2. SOME in the *Caterpillar* State.
3. SOME in the *Chrysalis* State.
4. AND some in the *Fly* State.

IT is likewise to be observed, that the several Species of them chuse different Places for their Residence

1. SOME frequent Woods.
2. SOME Plains and Heaths.
3. SOME Hedges and Thickets.
4. SOME Meads, and Mowing Grass.
5. AND others marshy wet Grounds, &c.

SEVERAL Kinds of *Caterpillars* come forth of the Egg in *July, August, September, and October*, remain in the *Caterpillar* State all the Winter, feed up in the Spring, then change into the *Chrysalis* State, and produce *Flies* a few Weeks afterwards.

OTHERS there are that continue in the Egg State all the Winter, are hatched in the Spring, and feed up in about 45 Days: then change into the *Chrysalis*, and produce their *Flies* in about 14 Days afterwards.

N. B. The Curious may be assured, that all these Observations are founded on real Facts, and not on any Supposition or Conjecture.

AND now, (as it seems exactly suitable to this Work) we shall beg Leave to close the Introduction with a short Description of the *Butterfly*, taken from a Poem called The UNIVERSE,

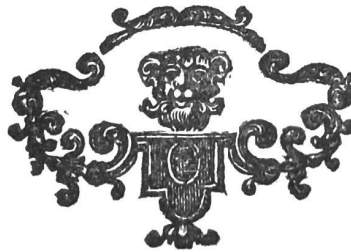
which was written some Years ago
By Mr. *Henry Baker*.

S E E, to the Sun the Butterfly displays
Its glittering Wings, and wantons in his Rays:
In Life exulting, o'er the Meadows flies,
Sips from each Flow'r, and breathes the vernal Skies.

Its splendid Plumes, in grateful Order, show
The various Glories of the painted Bow.
Where Love directs, a Libertine it roves,
And courts the fair ones thro' the verdant Groves.

How glorious now! How chang'd since Yesterday! }
When on the Ground, a crawling Worm it lay, }
Where ev'ry Foot might tread its Soul away! }
Who rais'd it thence? And bid it range the Skies?
Gave its rich Plumage, and its brilliant Dyes?

'Twas God:—Its God and thine, O Man, and He }
In this thy Fellow-Creature lets thee see, }
The wond'rous Change which is ordain'd for thee. }
Thou too shalt leave thy reptile Form behind, }
And mount the Skies, a pure ethereal Mind, }
There range among the Stars, all bright and unconfin'd. }



An Account of those *Places* where some GENTLEMEN
OF THE
AURELIAN SOCIETY

Have been used to collect *Caterpillars*, *Chrysalides* and *Flies*,

In the following Months of the Year.

M A R C H.

FROM the 15th to the 25th of this Month, a *Moth* called the *Orange Under-Wing*, may be taken in *Hornsey-Wood*, and in the Fields close to the Sides thereof. It flies in the Day-time, is an exceeding pretty *Moth*, and not to be met with in such Plenty any where else, that I know of. The *Oak-Beauty*, and some other *Moths* are bred in this Month, but are rarely met with in the *Fly State*.

A P R I L.

IN this Month a great Variety of *Caterpillars* and *Chrysalides* may be taken by the Method already described. The Banks about the *Chelsea Water-Works* and such-like Places are most likely to furnish the *Chrysalides* of the *Hawk Tribe*.

At the Roots of the *Elm*, *Lime*, *Willow*, and *Poplar Trees*, you will find in this *Month* a Variety of *Aureliæ*, which will produce their *Flies* in the *May* and *June* following.

Be now diligent to collect *Caterpillars* from all kinds of Growths, which, if taken Care of, will produce their *Flies* a few Months hence. Towards the End of this Month some *Flies* may be taken in the Day-time, and some *Moths* in the Evening.

M A Y.

THIS delightful Month brings forth in the Woods and Meads a surprising Variety of *Flies*. *Hornsey*, *Cain*, and *Tottenham* Woods; the Woods near *Southgate*; *Hanging Wood*, by *Charlton*, in *Kent*; *Oak of Honour Wood*, by *Dulwich*; *Comb Wood*, by *Kingstone*, in *Surrey*, and the Fields adjacent are all frequented by them in this Month in the Day-time as well as in the Evening; and if the Weather be fine you will never fail of Sport. Abundance of different *Caterpillars* are likewise to be found at this Time, such as those of the *Emperor-Moth*, of the *Lappit*, *Drinker*, *Egger*, and of Numbers of other Sorts.

J U N E.

THIS Month adds greatly to the Number of *Flies* bred in *May*.—From the 20th to the 30th are usually produced: in *Southgate*, *Comb*, and *Oak of Honour* Woods, the *Great Fritillary* and other fine *Butterflies*, with Plenty of *Moths*; all which may be taken in and near the Woods already mentioned.

J U L Y.

THE first ten Days in this Month are the Time for taking the *Purple Emperor-Butterfly* in *Comb Wood*; and the Fields adjacent can furnish such a charming Variety of *Moths* and *Butterflies*, that I do not know any Place where an *Aurelian* can spend a Week with more Satisfaction and Assurance of Success. As soon as the *Purple Emperor-Butterfly* begins to grow bad, the Time comes on when the second Breed of the *Swallow-tail Butterfly* is quite fresh, as also the *Great Tortoise-shell*, *Peacock Butterfly*, &c. About the Middle of this Month you may find the *Great Red Underwing* and the

Goat-Moth standing against the *Willow-Trees* which usually grow by the Sides of Ditches about the old Barge-House, *Rotherhith*, *Vauxhall*, and likewise in other Places. Seek also for the *Caterpillars* that produce the *Great Elephant* and *Gold-Spot Moths*: the most certain Places to find which, that I know of, are by the Sides of the Ditches in *Rotherhith* Marshes. The *Caterpillars* that produce the *Elephant Moth* feeds on the *White Ladies Bedstraw*, those of the *Gold-Spot* on the *Great Water-Grass*. I have taken the *Gold-Spot Chrysalis* and *Caterpillar* likewise in the low Grounds by *Vauxhall* in *Surry*.

A U G U S T.

AT the Beginning of this Month may be found amongst the *Willows*, about the Banks by the *Chelsea Water-Works*, the *Caterpillars* that produce the *Eyed-Willow* and *Poplar-Hawks*, also the *Buff-Tip-Puss*, &c. and from the 20th to the 30th the *Admirable Butterfly*, &c.

S E P T E M B E R.

IF this Month proves fine, a great many scarce *Moths* are to be taken in the Evenings, and the second Brood of the *Comma-Butterfly*, as well as some other *Flies*, present themselves. *Caterpillars* are likewise to be got in Plenty off the *Oak*, *Black-Thorn*, and a Variety of other Foods; and these, if taken Care of, will produce scarce and valuable *Flies* in the succeeding

Spring. From the End of this Month to the Middle of *March* there are but few *Flies* that frequent the Fields; altho'

I am of Opinion that some *Moths* are
bred in every Month of
the Year.



In order to oblige such persons as may be desirous
To make a COLLECTION of
MOTHS and BUTTERFLIES,

Though unacquainted with the Manner how,

It is judged proper to lay down the following

DIRECTIONS.

PROVIDE yourself with a Net made of Muschetto Gause, and in Shape like a Bat-fowling Net; let its Length be one Ell, the Width at Bottom three Quarters of a Yard, at Top half a Yard, and cut circular; this must be sewed to a Tape or Ferret, that it may be fastened to a couple of Hasle or other Sticks five Feet long each, the upper Part whereof should be bent circular to fit your Net.

You must likewise have a Stick of Hasle, or any other Wood, about five or six Feet in Length, wherewith to put the *Flies* and *Moths* on the Wing, by beating the Boughs, Hedges, Shrubs, or other Growths you are near, so that you may be better able to see and take them.

Furnish yourself likewise with Boxes of such Sizes as will go into your Pocket, let them be lined at the Top and Bottom with Cork; be also provided with a Pin-cushion, well stock'd with different Sorts of Pins.

These Things being in readiness, go into the Woods and Fields thereto adjacent (always placing yourself where the Sun has most Power, and the Wind the least) also into Chalk-pits, rough Grounds, Lanes, &c. In all which Places, if it is a fine Day, and a proper Season of the Year, you'll find Plenty of Sport.

When you have taken a *Fly* in your Net, lay the Net flat on the Ground, which will prevent the *Fly* from fluttering, then put the Fore Finger of your Left Hand under the *Fly*, and with the same Finger of your Right Hand give it a squeeze on the Back or Chest, and that will kill it; be careful, however, not to press too hard, least you damage the *Fly*. This done take a Pin and run it through the Body, betwixt the Wings, letting that Side be uppermost which is most beautiful; then stick it in your Box, and look for more Sport.

Having collected such a Number of *Flies* as you think proper, and being returned home, look into your Boxes, and observe which of them are fit to set: such as you find dead and not stiff are so.

Then (having prepared before-hand two or three Boards covered with Cork, of about ten Inches by sixteen in Size, or as you find most convenient to place your *Flies* in order on and extend their Wings in the best manner: by the Assistance of little Braces made by cutting a Card into Lengths or Slips, with a small Pin thrust through the End of each) proceed to manage them as follows:

Take a *Fly* out of your Box: see if the Pin be run through it perpendicularly: if so, stick it on one of your setting Boards, and with the Point of a Needle (which must be fixt into a small Stick, or what else you like best) extend one Wing leisurely, till such Time as the Point thereof is even with the Nose of the *Fly* you are setting. That done, fix one of your Card Braces gently on that Wing, to prevent its giving way; serve the other Wing in the same manner, and your *Fly* will appear extended as in the Paintings. Let the Braces remain on the Wings of *Butterflies* a Fortnight, on those of great *Moths* a Month. Take Notice, however, that a great Number of small *Moths* must be set in your Boxes in the Field, otherwise your labour will be lost; you'll presently know which these are, by observing them to be dead and almost stiff; so that it is proper to carry always about you a little Box of Card Braces for this Purpose.

The Way to preserve your Flies, after you have taken them from your Setting Board.

If you put them in Drawers, Boxes, or Frames with Glasses before them, its proper to get some Camphire, which inclose in small Muslin Bags, and fasten it to those Places where your *Flies* are; by this Means you will destroy those Insects that would otherwise injure your *Flies*; and by

renewing now and then this Method, I am convinced your *Flies* may be preserved several Years. The Months in *England* that produce the greatest Variety of *Flies*, are *April, May, June, July, and August*, especially of the *Butterfly* Kind; and for *Moths*, I am of Opinion, that there are many different Species produced all the Year round. The best Time to take the greatest Variety of *Moths*, is for one Hour after Sun-set, and the properest Places are in and by the Sides of Woods, Gardens, Green Lanes, &c. where with your Net you'll seldom fail of Sport. Thus much seemed necessary to say in Respect to taking *Insects* in the *Fly* State, and more would be superfluous.

But as it may be agreeable to some to breed the Flies themselves from the Caterpillars, I shall likewise give the best Instructions I am able for that Purpose.

IN the Spring and Autumn Season you may collect great Variety of *Caterpillars*, by spreading a Sheet under *Oak-Trees*: then beating the Boughs, many *Caterpillars* will fall, which if taken Care of, will produce scarce and valuable *Moths*; you may also obtain by the same Means *Caterpillars* from the *Black-Thorn, White-Thorn, Bramble, Chickweed, Willow*, and many other Growths.

When you have collected a Number of *Caterpillars*, and remark'd the Food you found them on, put them into Boxes, which prepare as follows: Take a Deal or Wainscot Box, cut a large Square out of the Top and Bottom, cover the Place cut out with a Piece of Crape Hatband, glewing the same all round to prevent any Escape. This done, put your *Caterpillars* into the Box, with some of the same Food you found them on, giving them, if you can, fresh every Day. Here you'll find them feed and thrive; and after changing their Skins (some five and some six Times) they will go into the *Aurelia* State, and there remain for a certain Time, some much longer than others; but in about fourteen Days from this last change you may look into your Boxes, to see if you have any *Flies* bred; remembring that out of such as are produced in this manner, much more perfect *Flies* may be chosen, than any that can be caught; and as several Sorts of *Caterpillars* go down into the Earth, and there change to *Chrysalis*, some Earth must be put into the Boxes for that Purpose, at least two or three Inches deep. Never take the *Chrysalides* out of the Earth, nor disturb them till the *Flies* are bred.

There is yet another Thing proper to be known, in order to compleat your Collection. A great Number of *Caterpillars* go into the Earth, and there change to *Aureliæ*, whose *Moths* are seldom upon the Wing till Night; these *Aureliæ* are to be got by digging with such a Trowel as the *Bricklayers* use, about the Roots of Trees, such as *Oaks*, *Elms*, *Limes*, *Poplars*, *Willows*, &c. also by the Sides of Walls and Pales. The usual Time to dig for them is from *September* to *March*.

When you have obtain'd a Number of *Aureliæ* by digging, you must provide for them thus: Get some Earth and scowering Sand, mix them well together, and put the Mixture into some large Garden Pots; lay your *Aureliæ* thereon, covering them with Moss; then sew a Piece of Crape Hatband round a small Hoop, and put such a Cover on each Pot, in order to receive the *Flies* when bred, which will usually be in the Months of *March*, *April*, *May*, and *June*. I have always kept my Pots of *Aureliæ* out in the open Air, and the *Flies* have been produced with great Success; however it may be proper to place a Piece of Board over each Pot, to prevent unforeseen Accidents.

Those who think proper to put these Directions in Practice, will, I am convinced, in a short Time, be possessed of a great Number of valuable *Flies*. What Food the *Caterpillars* mentioned in this Work chuse to feed on, the Time when they change into *Aureliæ*, and when they issue forth in the *Fly* State, may be seen in the Description of each Plate.

ERRATA.

Page.	Num.	Read.	Instead of
8	XVII	Chrysalis	Chrysalisis
11	VI	nigra	alba
18	II	Cl. 3. S. A. 2.	Cl. 3. S. E. 1.
20	VII	form	from
23	XIII	to	o
49	I	Sect. a.	
51	V	Cl. 2. S. a. 5.	Cl. 2. S. a. 2.

52	I	4 The Great } Fox-tail Grass.}	
53	IV	Cl. 2. S. b. 4.	Cl. 2. S. <i>a.</i> 4.
62	III	Cl. 1. S. <i>a.</i> 3.	Cl. 2. S. <i>a.</i> 4.
63		of Butterflies	of Caterpillars

On the Plates.

38	I	Cl. 1. S. b. 1.	Cl. 1. S. <i>a.</i> 1.
38	II	Cl. 1. S. b. 2.	Cl. 1. S. <i>a.</i> 2.
42	I	Cl. 2. S. <i>a.</i> 1.	Cl. 2. S. 1. b. 1.
42	II	L. I. Ch. III.	L. I. Ch. II.
62	II	Cl. 1. S. <i>a.</i> 2.	Cl. 1. S. <i>a.</i> 1.

A Synoptical T A B L E of the *Moths* and *Butterflies* contained in this Work.

CATERPILLARS.

Producing Moths.

Having circular Rings or Joints.

Naked.

Without Protuberances.

THE great yellow Underwing. The Wild Arrach. The Angle Shades. The Pease Blossom. The small green Oak Moth. The small Ermine Moth. The Broom Moth. The sallow Moth. The Codling Moth. The Province Rose Moth. The Scallop-wing'd Oak Moth. The Black-thorn Moth. The green Moth with silver Lines. The Ranunculus Moth. The Water-Betony Moth. The small Elephant Moth. The Gold-spot Moth. The Sword-grass Moth.

With Horns on the Tail.

The Jessamine-Hawk Moth. The Unicorn, or Bindweed-Hawk Moth. The Privet-Hawk. The Olive-shades, or Lime-Hawk. The Eyed Willow-Hawk. The Poplar-Hawk. The Ladies Bedstraw, or Elephant Moth.

With one Protuberance on the Rump.

The Ealing's Glory.

With two or more Protuber.

The Ozier, or Pebble Moth.

With Shoulders rising into an Angle.

The Puss Moth. The Wild Rose Moth.

Having little Hair.

Without Protuberances.

The Goat Moth. The Emperor Moth. The Plumb-Tree Moth.

With Protuberances.

The Willow Red-Underwing.

Having much Hair.

Without Protuberances.

The Great Tyger. The Cream-Spot Tyger. The Scarlet Tyger. The Black Arches. The Great Ermine Moth. The Spotted Buff Moth. The Gipsy Moth. The Buff Tip. The White Satin Moth. The Lacquey. The Great Egger. The Grass Egger. The December Moth. The Spotted Red and White Underwing. The Wood Tyger. The Mother of Pearl Moth. The Yellow July Oak Moth. The small Egger. The Fox-coloured Moth.

Having Rings of different Colours.

The Cinnabar Moth.

Having indented Marks.

The Bramble Moth.

Having fleshy Protuberances on the Head, Back, or Tail.

The Lappit-Moth. The Drinker. The Yellow Tail. The Dagger Moth. The Wild Pine Tree Lappit-Moth.

Having many Tufts of Hair.

The Black Tussock. The Yellow Tussock. The Red Spot Tussock. The Orange Tussock. The Nut-Tree Tussock. The Sycamore Tussock.

Half Loopers.

Having 8 Feet, 2 Holders, 6 Claws, and with Protuberances.

The Crimson Underwing.

Having 4 Feet, 2 Holders, 6 Claws. Hairy.

The Silver Y Moth.

Loopers.

Naked.

Without Protuberances.

The Brindled Beauty. The July Arrach Moth. The Mottled Umber. The July Sallow Moth. The Buff Argus. The Green Broom Moth. The Lime Moss Moth.

With Protuberances.

The Spotted Elm Moth. The Swallow-Tail Moth. The October Moth. The Brimstone Moth. The Scallop-winged Moth. The Richmond Beauty. The Hawthorn Moth.

Hairy.

Without Protuberances.

The large Magpye, or Curran Moth. The Gooseberry Moth.

Having Tufts of Hair.

The Brindle Moth.

Shaped like Wood-lice.

Small Oak Egger-Moth.

Producing Moths whose Generation is unknown.

Glory of Kent. Cleifden Nonpareil.

Producing Flies between the Moth and Butterfly.

The Burnet Moth.

Producing Butterflies.

Smooth.

Without Protuberances.

Swallow-Tail Butterfly. Brimstone Butterfly.

Having little Hair.

Producing round-winged Butterflies.

White Butterfly with black Veins. Large white Garden Butterfly. Small white Garden Butterfly. White Butterfly with green Veins. Orange Tip, or Lady of the Woods.

Producing scallop-wing'd Butterflies.

The Marmoris, or Marble Butterfly. The Meadow Brown. The Great Argus, and Wood Argus.

Producing Butterflies with large Heads and Bodies.

The Grizzled Butterfly.

Armed with Spikes.

Whose Chrysalis hangs by the Tail.

The Admirable Butterfly. The Peacock Butterfly. The painted Lady. The great Tortoise-shell. The Comma-Butterfly. The Great Fritillary. The Plantain Fritillary. The Heath Fritillary. The Willow Butterfly. The small Fritillary. The great Fritillary, with silver Spots.

Shaped like Wood-lice.

The Purple Hair-streak Butterfly. The Brown Hair-streak. The green Butterfly.

Producing Butterflies whose Generation is yet unknown.

The Blue Argus Butterfly. The Purple High-Flyer, or Emperor of the Woods.

T H E
NAMES of the Subscribers to, and Encouragers of,
Mr. WILKES's Attempt towards a
NATURAL HISTORY of
English MOTHS and
BUTTERFLIES.

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The Right Honourable the Lady Betty Germain.

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Mr. Thomas Grace.

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The Most Reverend Father in God Thomas, Lord Archbishop of
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The Reverend Mr. Thomas Manningham.

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Mr. Daniel Marshal.

Mr. Miller.

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The Reverend Mr. John Newton, at Gloucester.

O.

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R.

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The Right Honourable the Lord Rolle.
The Royal Society of London.

S.

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The Right Honourable the Countess of Stamford.
The Right Honourable the Lady Diana Spencer.
The Right Honourable the Lady Elizabeth Spencer.
The Right Honourable the Lady Smithson.
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Sir Hans Sloane, Baronet.
John Archer Shish, Esq;
Mrs. Smith, at Oxford.

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The Honourable Mrs. Trevor.
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Doctor Christopher Jacob Trew, at Norimburg.
The Honourable Mr. Talbot.
James Tilson, Esq;
Miss Tasburgh.

W.

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Ralph Willett, Esq;

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Horace Walpole Esq; of Arlington-Street.
Mr. William Wells.
Mr. Wollaston.
Mr. Woodhouse.
Mr. Whittle.
Mr. Ware.

Y.

The Honourable Mr. Yelverton, 3 Books.





B O O K I.

P A R T I.

The CATERPILLARS, described in this Book, are such as produce Moths; of which, as there is a great Variety, 'twas thought proper to range them under the following Heads or Chapters.

C H A P. I.

THE CATERPILLARS belonging to this Chapter have six Hooks or Claws before, eight Feet in the middle, and some have, and some have not two Holders behind. Their Bodies are composed of twelve circular Joints, or Rings, conjoined by a common Membrane, which they are capable of contracting or expanding. When they would move along, they dilate the muscular Skin that separates the first Ring from the next; they advance the first Ring to a certain Distance, and then, by contracting and expanding the Skin of that Part, they move the second Ring: the same Method moves the Third, and so the whole Body marches in Succession: the Motion of their Feet and Claws corresponding with and assisting the progressive Motion of the Rings. *Vide Spectacle de la Nature, Dial. 1.*

The CATERPILLARS under this Head may likewise be divided into three Classes:

- I. NAKED or without HAIR.
- II. Having little HAIR.
- III. Having much HAIR.

The First CLASS, or the NAKED and without HAIR, may be subdivided in the following Manner.

C L A S S I.

Such CATERPILLARS whose Bodies are SMOOTH.

S E C T . A.

Without Risings, or Protuberances

L. 1. CH. 1.

I.

CL. 1. S. A. 1.

THE GREAT YELLOW-UNDERWING MOTH. The CATERPILLAR of this MOTH lives through the Winter, and comes to its full Growth in *May*: It is found in Gardens, and other Places, and usually feeds in the Night, as many other of the NAKED CATERPILLARS do. Indeed, were they to do otherwise, they would soon be devoured by the Birds, which are much fonder of these SMOOTH CATERPILLARS than of the HAIRY ones. In the Day, they hide themselves within the Earth, and in other secret Recesses. They feed on the Leaves of the *Stock-July* and other Flowers, as also on Grass, &c. They put on the *Chrysalis* Form, within the Earth, in *May*, and the *Moths* are bred in *June*. Great Plenty of this *Moth*, and other Species of the YELLOW-UNDERWING, were taken feeding on the Honey Dew, on the Limes and other Trees, in the Gardens of *John Philips*, Esq; at *Layton* in *Essex*; they were discovered, by the Help of a Candle and Lanthorn, from Twelve o'Clock at Night till Two in the Morning; and were so fearless, that they would suffer one to take them with the Hand. At the same Time the ANGLE-SHADE and several other *Moths* were taken.

See *Lister* on *Godartius*, P. 52. N. 41. *Albin*, P. 72. *Reaumur*, Pl. 14. Pag. 342. vol. 1.



L. I. ch. 1.

Cl. 1. S. A. 1.

The Double Stock July-Flower.

Leucoium incanum, majus, variegatum, flore pleno. C. B. Pin. 201.

L. 1. Ch. 1.

II.

CL. 1. S. A. 2.

THE WILD ARRACH MOTH, Mr. Rosel says, The *Caterpillar* feeds on the WILD ARRACH; that it changes to a *Chrysalis*, within the Earth, in *September*, and that the *Moth* appears at the End of *May* following. I

once took some of these *Caterpillars* feeding on the *Arrach* in *September*, but had not the good Fortune to breed any Flies from them. See *Rosel*, Cl. 2. Tab. 32.



L. I. ch. I.

Cl. 1. S. A. 2.

The Wild Arrach in Flower.

Atriplex folio deltoide. Petiv. Herb.

L. 1. CH. 1.

III.

CL. 1. S. A. 3.

THE ANGLE-SHADES, a *Moth*. It is advisable to look for the *Caterpillar* of this Fly on Nettles, altho' some feed on Fruit Trees, Bramble Bushes, &c. It goes into the *Chrysalis* State at the End of *June*, in a Web mixed with Earth, which it forms upon the Ground; and thus it remains till the Beginning of *September*, when the *Moth* is produced. There is an early Breed of this *Moth* in *May*, but, as I have not had the *Caterpillar* in the Spring, I cannot relate its Progression. The Fly may be taken in the Evenings, in Gardens, feeding on the Flowers; but I have not met with them very frequent.

See *Godartius*, Part 1. Tab. 56. *Lister* on *Godartius* P. 54. N. 44. *Albin*, Pl. 13. *Reaumur*, Pl. 8. P. 258. Pl. 14. P. 342. Vol. 1.



L. L. ch. 1.

Cl. 1. S. A. 3.

The Great Mogul Plum.

Prunus fructu magno ovato rubente. Tourn.

L. 1. CH. 1.

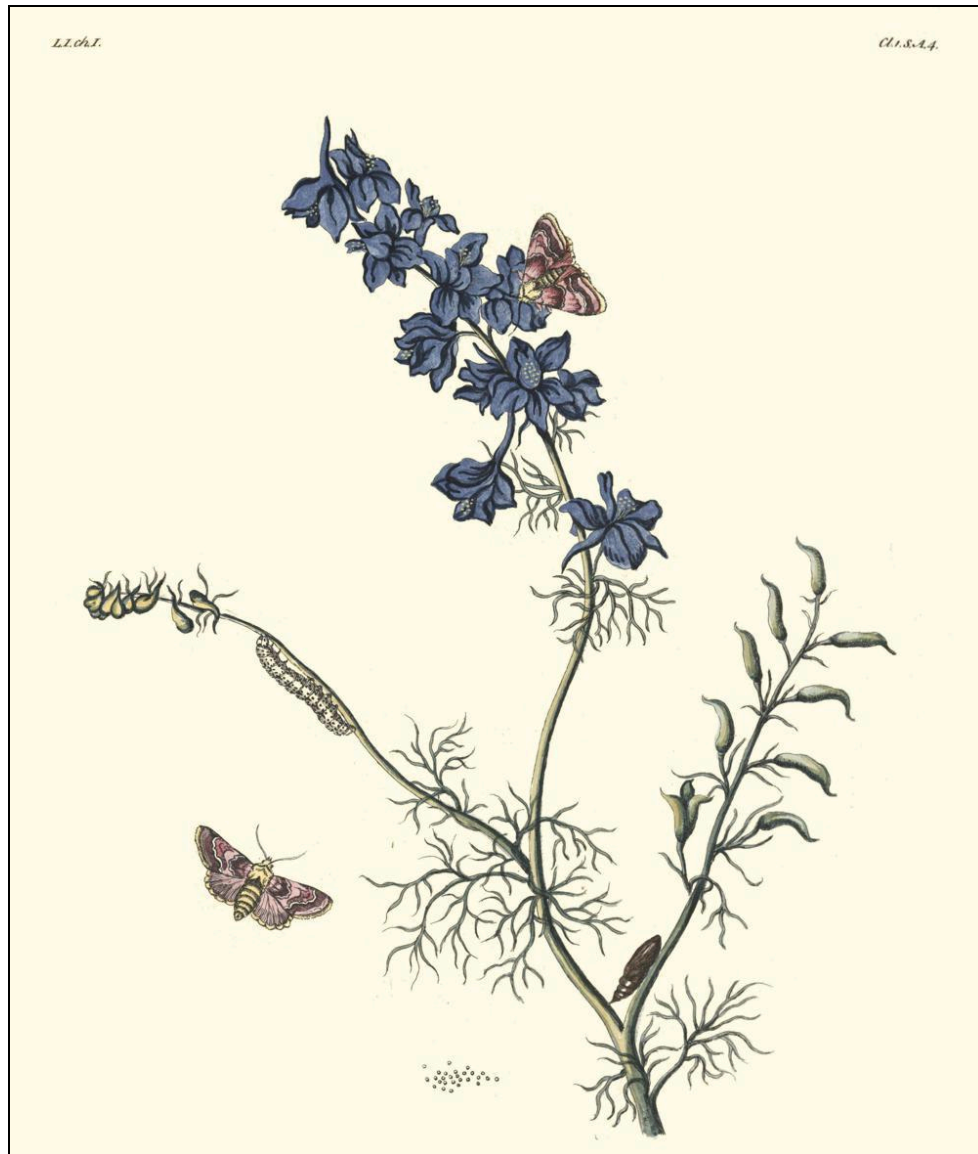
IV.

CL. 1. S. A. 4.

THE PEASE-BLOSSOM, a *Moth*. Its *Caterpillars* are reported by *Merian*, and *Rosel*, to feed upon the wild Lark-Spur that grows amongst Corn. They are full fed at the Beginning of *August*, when they change to the

Chrysalis, but appear not in the *Moth* State till the *June* following. This Fly has been bred in *England* by the Honourable Mrs. *Walters*, and by *Nathaniel Oldham*, Esq; but, at present, is very rare.

See *Merian*, Vol. 1. Ch. 40. *Rosel*, Cl. 2. Tab. 12.



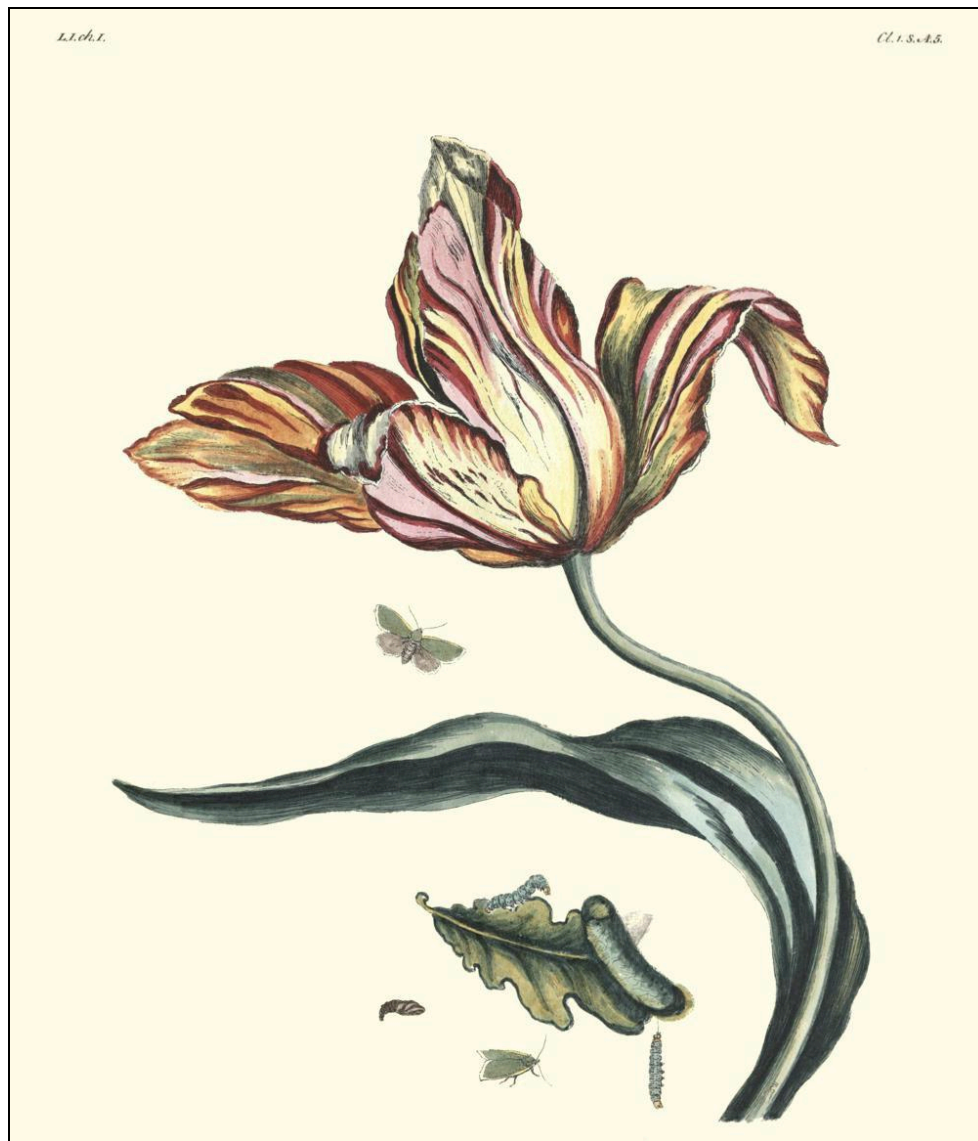
L. I. ch. I.

Cl. 1. S. A. 4.

The Lark-Spur.

Delphinium, hortense, flore majore & multiplici cœruleo. Tourn.

THE SMALL GREEN OAK MOTH. The *Caterpillar* feeds on the Oak, and sometimes such Numbers of them are produced at the Beginning of *May*, that they, in a Manner, strip the Oak-trees of their Leaves. They are enabled, by the Help of a Spining, which they fasten after the Manner of the Spider, to let themselves down from one Bough to another, and from thence to the Ground, if they have a mind. They change into the *Chrysalis* State in a curled Form within the Oak-leaves, (which they wrap over them, and fasten together by their Web) and, after remaining so for about fourteen Days, the Fly is bred. If you beat the Branches of the Oak-trees you may take Plenty of this *Moth* in the Day-time.



L. 1. ch. 1.

Cl. 1. S. A. 5.

The Yellow Strip'd Tulip.

Tulipa hortensis flore luteo variegato.

L. 1. CH. 1.

VI.

CL. 1. S. A. 6.

THE SMALL ERMINE MOTH. The *Caterpillar* that produces this *Moth* is as common as those just mentioned. I have seen a Hedge-row of White-thorn, at the End of *May* and at the Beginning of *June*, the Leaves of which have been eaten almost up by these *Caterpillars*. They feed also on Black-thorn, Fruit-trees, &c. when they are preparing to change to *Chrysalis*, they get together in Numbers, and spin a large transparent Web, within which they fasten themselves by the Tail, and undergo their Change. They remain in this Condition for about three Weeks: The *Moths* come forth in *June* and *July*, and are very common about most Hedges.

See *Albin*, P. 70. *Reaumur*, Pl. 12. Pag. 208. Vol. 2.



L. I. ch. I.

Cl. 1. S. A. 6.

The Orange-Peach, with its Blossom.

Persica, dura carne buxea. C. B. Pin.

L. 1. CH. 1.

VII.

CL. 1. S. A. 7.

THE BROOM-MOTH. The Food of this *Caterpillar* is the Leaves of Broom, on which I found Plenty of them, from the 20th to the 30th of *September*, near *Shooter's Hill* in *Kent*; they were of the Bigness express'd in the Plate. The Green ones produced male Flies. They all went into the

Earth at the Beginning of *October*, and, at the Beginning of *May* following, the *Moths* were bred. I never could take any of them in the Fly-state.

See *Rosel*, Cl. 2. Tab. 52.



L. 1. ch. 1.

Cl. 1. S. A. 7.

The Broom, in Flower and Seed.

Genista angulosa Trifolia. I. B. 1388.

L. 1. CH. 1.

VIII.

CL. 1. S. A. 8.

THE SALLOW MOTH. The *Caterpillar* was fed by Mr. *Dandridge* on the common Sallow, until it changed to a *Chrysalis* within the Earth, which was at the Beginning of *June*. The *Moth* was produced about the Middle of *September*. These *Moths* are not common; for, in all my Practice, I have not been able to take above three or four, and that was in the Evening, near *Shooter's Hill*, in the Month of *September*. The *Caterpillars* are likewise very scarce.

See *Albin*, Pl. 33.



L. I. ch. I.

Cl. 1. S. A. 8.

Common Sallow in Flower.

Salix Cutifolio rotundo. C. B. Pin.

L. 1. CH. 1.

IX.

CL. 1. S. A. 9.

THE CODLING MOTH. Mr. *Rosel* says, The *Caterpillar* of this *Moth* feeds on the Kernels of Apples and Pears. The *Moth* lays her Eggs at the End of *June*: Eight Days after the *Caterpillars* are bred; and, when full fed, they go out of the Fruit and change to the *Chrysalis* under the Bark of the Tree. They remain there all the Winter, and the *Moths* are bred in *June* following. I have, myself, found Plenty of these Flies sticking on the Bark of the Oak-trees, in *Richmond* Park, at the End of *June*, in the Day-time.

See *Rosel*, Cl. 4. Tab. 13.



L. I. ch. I.

Cl. I. S. A. 9.

The Codling-tree with its Blossom.

Pomum Coctile. Raii Hist.

L. 1. CH. 1.

X.

CL. 1. S. A. 10.

THE PROVENCE ROSE MOTH. Rose Leaves are the Food of this *Caterpillar*. In the Middle of *May* I had some of them that changed to

the *Chrysalis* folded in the Leaves; at the Beginning of *June* the *Moths* were bred. They may be taken in Gardens in the Evening.

See *Rosel*, Cl. 4. Tab. 9.



L. 1. ch. 1.

Cl. 1. S. A. 10.

The Moss Provence Rose.

Rosa, Provincialis spinosissima, pedunculo muscoso. Boerh. Ind. alter.

L. 1. Ch. 1.

XI.

CL. 1. S. A. 11.

THE SCOLLOP-WINGED OAK MOTH. I took the *Caterpillar* of this *Moth*, feeding on Oak-leaves, about the 10th of *June*, at which Time it went into the *Chrysalis*, spinning itself up in a Leaf; and the *Moth* was bred on the 7th Day of *July*. Both *Moth* and *Caterpillar* are very scarce.

See *Rosel*, Cl. 2. Tab. 63.



L. I. ch. I.

Cl. 1. S. A. 11.

The Yellow Rose.

Rosa, Lutea, multiplex. C. B. Pin. 4831.

L. 1. CH. 1.

XII.

CL. 1. S. A. 12.

THE BLACK-THORN MOTH, or, *Figure of 8*. The *Caterpillar* is easily found, feeding on Black Thorn, White Thorn, Crab-tree, &c. towards the End of *May*, at which Time it spins a hard Case against its Food, and changes to a *Chrysalis*; the *Moth* is bred about the End of *August*, and is not commonly taken.

See *Albin*, P. 13. *Rosel*, Cl. 2. Tab. 16. *Reaumur*, Pl. 18. Pag. 342. Vol. 1.



L. 1. ch. 1.

Cl. 1. S. A. 12.

The common Periwinkle.

Pervinca vulgaris angustifolia, flore pleno saturate purpureo. Tourn.

L. 1. CH. 1.

XIII.

CL. 1. S. A. 13.

THE GREEN MOTH WITH SILVER LINES. I fed its *Caterpillar* with Oak-leaves, on which it had been found. It work'd itself up in a dark brown Case on the Back-side of one of the Leaves, and changed into a *Chrysalis* the 27th of *September*; the *Moth* appeared the 20th of *May* following, and is difficult to meet with.

See *Albin*, P. 31.



L. 1. ch. 1.

Cl. 1. S. A. 13.

The Bruxells Apricock.

Malus Armeniaca, Bruxells vulgo dicta.

L. 1. CH. 1.

XIV.

CL. 1. S. A. 14.

THE RANUNCULUS MOTH. The *Caterpillars* represented in this Plate were found feeding on the *Ranunculus* Flower about the Middle of *June*, at which Time they were full fed, and soon changed to the

Chrysalis within a Spining or Web; and the *Moths* came out at the Beginning of *September*. The *Caterpillar* and *Moth* are both very rare.

See *Rosel*, Cl. 2. Tab. 53, 54.



L. L. ch. I.

Cl. 1. S. A. 14.

The double Orange Ranunculus.

Ranunculus hortensis, flore aurantio pleno.

THE WATER-BETONY MOTH. The *Caterpillars* are sociable, and feed together. They may be found, towards the End of *May*, full fed, on the *Water-Betony*; as also on the *Mullein*, which is their proper Food. They go into the Earth about that Time, and change to the *Chrysalis*, within a Case composed of Earth and a webby Matter, which they spin together. The *Moth* appears in *March* following; but it is not commonly taken in its Fly-state.

See *Raii* Hist. Insect. P. 168. N. 25. *Albin*, Pl. 13. *Rosel*, Cl. 2. Tab. 22. *Reaumur*, Pl. 43. P. 602.



L. 1. ch. 1.

Cl. 1. S. A. 15.

The Water-Betony; or Water-Figwort.

Betonica Aquatica, Ger. 579. *Scrophularia Aquatica major*. C. B. Pin. 235.

L. 1. Ch. 1.

XVI.

CL. 1. S. A. 16.

THE SMALL ELEPHANT MOTH. Mr. *Rosel* informs us, That the *Caterpillar* is to be found on the Grass in Meadows, in the Month of

August; but whether that be its favourite Food he does not know. It makes a Spining, and changes to a *Chrysalis* towards the End of that Month. The *Moth* is bred about the End of *May*.

In the Year 1743, on the 28th of *May*, I took several of these *Moths*, in the Evening (after Sun-set) as they were hovering about, and feeding on a Flower called *The Batchelors Button*, by the Side of *Osterly Wood*, near *Brentford*. They are very swift in flight, and therefore must be taken as soon as they approach the Flower. The Females layed their Eggs freely, and, after seven Days, the *Caterpillars* were hatch'd, to the Number of about 300. I took the utmost Pains to raise them; but, for want of their proper Food, they all died.

See *Rosel*, Cl. 1. Tab. 5.



L. 1. ch. 1.

Cl. 1. S. A. 16.

The Gelder-Rose.

Opulus, Flore Globoso. Tourn. Inst. 607.

L. 1. CH. 1.

XVII.

CL. 1. S. A. 17.

THE GOLD-SPOT MOTH. The *Caterpillar*, whence this *Moth* is produced, may be found about the End of *July*, full fed, in the Marshes about *Rotherhith, Vauxhall*, and such like Places. It feeds on the Leaves of

the great Water-grass, &c. but the *Caterpillar* is not so easily found as the Spining which it makes to contain its *Chrysalis*; for that spining, being of a whitish Colour, may be seen at a good Distance, in Shape and Manner as described in the Plate. It lies in the *Chrysalis* State about three Weeks, and the *Moth* is bred from the Middle to the End of *August*; but it is seldom taken in the Fly-state.



L. I. ch. 1.

Cl. 1. S. A. 17.

Purple Flowered Ketmia.

Ketmia, Syrorum, Flore purpureo violaceo. Tourn. Inst.

THE SWORD-GRASS MOTH. Mr. *Rosel* informs us, That the *Caterpillar* of this Fly feeds on the *Orache*; that, in *June*, it changed to a *Chrysalis* within the Earth, in which State it remained three Weeks, and then the *Moth* was bred. I once took one of these *Caterpillars*, full grown, feeding on the *Sword-grass* in the Marshes at *Rotherhith*; but that was in the Month of *August*, and the *Moth* was bred with me at the End of *September*. Both *Caterpillar* and *Moth* are very scarce.



The Persian Iris.

Xiphium Persicum, præcox, flore variegato. Tourn.

SECT. B.

CATERPILLARS *having Horns on their Tails.*

L. 1. CH. 1.

I.

CL. 1. S. B. 1.

THE JESSAMINE-HAWK MOTH. This *Moth* was bred in *England* by Mr. *Dandridge*, who took the *Caterpillar*, at *Hampton-Court*, full fed, on the *Jessamine*, about the Middle of *July*; at which Time it went into the Ground, and changed to a *Chrysalis*, and the *Moth* came forth at the End of *October*. This Gentleman had also five or six *Caterpillars*, of the same Kind, brought to him, that were taken upon an Elder-tree in *Goodman's Fields*, which likewise changed to *Chrysalis* in the Ground in *July*; and the *Moths* were also bred in *October*: From whence I conclude that to be the Time when the Fly is usually bred.

See *Albin*, P. 6. *Reaumur*, Pl. 14. Pag. 342. Vol. 1.



L. 1. ch. 1.

Cl. 1. S. B. 1.

The Jessamine in Flower.

Jasminum vulgatum flore albo. C. B. Pin. 397.

L. 1. CH. 1.

II.

CL. 1. S. B. 2.

THE UNICORN, or BINDWEED-HAWK MOTH. Mr. Rosel reports this *Caterpillar* to feed in Corn-fields, on the Green-leaves of the Corn. At the End of *July* it was full fed, and went into the Earth in order to

change to its *Chrysalis* State; and the *Moth* was bred in *June* following. *Merian's* Account of this Fly is, That the *Caterpillar* feeds on the Roots of the *Lesser Purple-Bindweed*, which grows amongst Corn; that it went into the Ground, and changed to a *Chrysalis* at the End of *July*, and that the *Moth* was bred in *September*.

I have not yet had the good Fortune to meet with this *Caterpillar*, but have seen several of the *Moths* that have been bred and taken in *England*. The Dung of this *Caterpillar* is of a very extraordinary Size and Figure; as well as some others that are expressed in the Plates.

See *Rosel*, Cl. 1. Tab. 7. *Merian*, Chap. 25. *Reaumur*, Pl. 24. Pag. 322. Vol. 2.



L. I. ch. I.

Cl. I. S. B. 2.

The Small Bindweed.

Convolvulus minor vulgaris. Park. 171.



L. 1. ch. 1.

Cl. 1. S. B. 2.

L. 1. CH. 1.

III.

CL. 1. S. B. 3.

THE PRIVET-HAWK MOTH. The *Caterpillars* feed on *Privet*, *Lilac*, *Holly*, &c. and are to be found in Nurseries and Gardens. Look on the Ground under such Trees; and if you discover any of their Dung, that will direct you to the *Caterpillars* on the Trees above. They go into the Ground, and change to the *Chrysalis* State in *August* and the *Moth* appears in *June* following. 'Tis not very common to take the *Moth* in the Fly-state.

See *Raii Hist. Insect.* P. 144. N. 1. *Moufet.* P. 91. N. 5. P. 182. Ch. 2.
Albin, Pl. 7 *Reaumur*, Pl. 20. P. 282. Vol. 2.



L. I. ch. 1.

Cl. 1. S. B. 3.

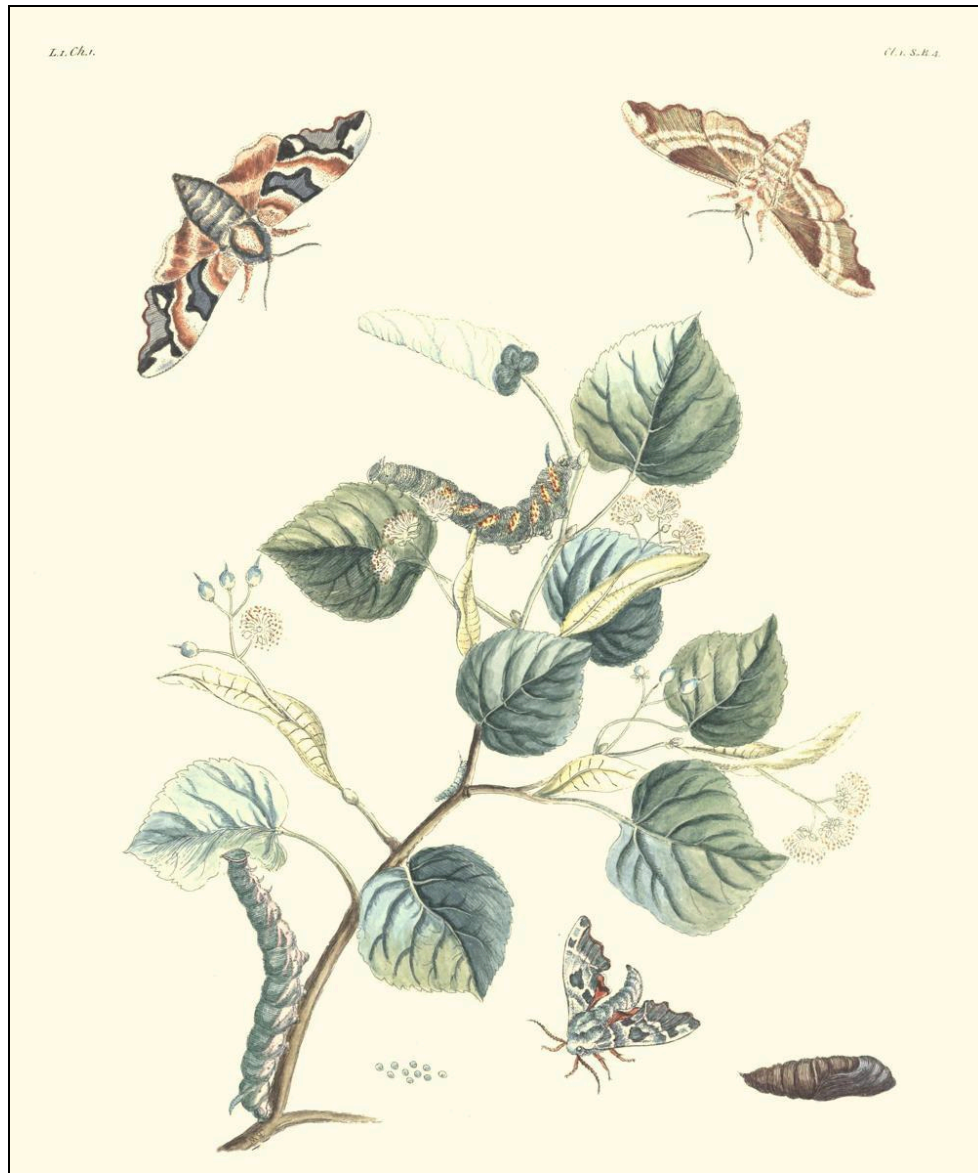
The Evergreen-Privet in Flower.

*Ligustrum; foliis majoribus, & magis acuminatis, toto anno folio
 retinens.*

Pluk. Alm.

THE OLIVE-SHADES, or LIME-HAWK MOTH. This *Caterpillar* is to be got by beating the Boughs of the *Lime* and *Elm* Trees, in the Month of *August*; at which Time it goes into the Earth, and there changes to a *Chrysalis*, and the *Moth* is bred in *May*. By looking carefully about the Bodies of *Lime* and *Elm* Trees, in *May*, these and other *Moths* may be found: For, if the Flies are bred in the Morning, or any Part of the Day, they never offer to fly till Night. You may likewise get these *Moths*, and many other Sorts, in the *Chrysalis* State, by digging with a Trowel about the Roots of the Trees the *Caterpillars* feed on; and the best Time to dig for the Hawk Kind is the Month of *April*. This Fly is seldom taken on the Wing.

See *Merian*, Vol. 2. Ch. 24. *Albin*, Pl. 10. *Rosel*, Cl. 1. Tab. 11.



L. 1. ch. 1.

Cl. 1. S. B. 4.

The small leaved Lime-tree in Flower.

Tilia fæmina, Folio minore. C. B. Pin.

L. 1. CH. 1.

V.

CL. 1. S. B. 5.

THE EYED WILLOW-HAWK MOTH. The *Caterpillars* are to be found on the Willows that grow about the *Chelsea* Water-works, and in most such like Places. At the End of *August*, I have also found them in

Gardens feeding on the Leaves of Apple-trees. They put on the *Chrysalis* Form within the Earth in *September*, and the *Moths* are produced in *May*. The *Chrysalis* and *Moth* may be got at the same Time as directed for the Lime-Hawk.

See *Lister's Godart.* N. 24. *Merian.* Vol. 2. Ch. 37. *Raii.* P. 148. N. 2. *Moufet.* P. 91. N. 6. *Albin,* Pl. 8.



L. l. ch. 1.

Cl. 1. S. B. 5.

The Crack Willow.

Salix, folio longo, latoque, splendente, Fragilis. *Raii.* Syn.

THE POPLAR HAWK MOTH. The *Caterpillar* of this *Moth* feeds on the Leaves of the *Poplar* Tree, as also on the *Willow*, and is to be found at the same Times and Places as the *Willow-Hawk Caterpillar*. It becomes a *Chrysalis* in the Ground in *September*, and the *Moth* appears in *May*. It is to be found in the *Chrysalis* and Fly-state as the before mentioned. All the Hawk-tribe fly very swiftly, and I am inclined to believe feed in wet marshy Places, by reason I have never taken them on the Wing in Places where I have always met with Plenty of other Moths.

See *Albin*, Pl. 57.



L. 1. ch. 1.

Cl. 1. S. B. 6.

The Black Poplar-tree.

Populus nigra, minoribus foliis. C. B. Pin.

L. 1. CH. 1.

VII.

CL. 1. S. B. 7.

THE LADIES BEDSTRAW, or ELEPHANT MOTH. The *Caterpillars* are to be found in the Ditches in *Rotherhith* Marshes, and also by the *New River* Side, near *Hornsey*. They feed on the *White Ladies*

Bedstraw, and are full grown at the End of *July*; at which Time they usually go into the *Chrysalis* State, by making a light Spining amongst their Food, where they remain till the End of the following *May*: At which Time the *Moth* is bred; but very seldom taken in the Fly-state. Mr. *Albin* has mentioned something very remarkable of this *Caterpillar*, viz. its Dexterity in Swimming: For, feeding commonly in or near the Water, if at any time it happens to fall in, it turns itself on its Back, and swims, with its Head and Tail turned together, till it gets hold on some Part of the Plant, by which it helps itself up again. It lies in the Egg-state about nine Days.

See *Raii Hist. Insect. P. 145. N. 2. Albin, Pl. 9.*



L. I. ch. 1.

Cl. 1. S. B. 7.

The White Ladies Bedstraw.

Gallium Album. Ger. 967.

SECT. C.

CATERPILLARS *having one Protuberance on the Rump, with indented Markings.*

L. 1. CH. 1.

I.

CL. 1. S. C. 1.

EALING'S GLORY, a MOTH. Its *Caterpillar* feeds on *Black Thorn, White Thorn, Crab-tree, &c.* becomes a *Chrysalis* in the Earth in *May*, and the *Moth* appears in *September*. *Charles Lockyer, Esq;* has bred great Numbers of this *Moth* at his House at *Ealing*; but I have heard of no body else who has met with them in any Plenty.

See *Albin*, Pl. 14. *Rosel*, Cl. 11. Tab. 33.



L. I. ch. I.

Cl. 1. S. C. 1.

The Black Thorn.

Prunus Sylvestris.

SECT. D.

CATERPILLARS *having two or more* Protuberances.

L. 1. CH. 1.

I.

CL. 1. S. D. 1.

THE OZIER, or PEBBLE MOTH. Mr. *Dandridge* found the *Caterpillar* feeding on the *Dutch Willow*, in the Marshes near *Rotherhith*, about the Middle of *July*; it changed to a *Chrysalis* within the Leaves spun together; and at the Beginning of *August* came forth the *Moth*. This fly is very scarce.

See *Albin*, Pl. 14. *Lister's Godart*. N. 21. *Rosel*, Cl. 11. Tab. 20. *Reaumur*, Pl. 22. P. 282. Vol. 2.



L. 1. ch. 1.

Cl. 1. S. D. 1.

The Ozier.

Salix, vulgaris, alba, arborescens. C. B. Pin.

SECT. E.

CATERPILLARS *whose Shoulders rise into an angular Figure, having no Holders behind.*

L. 1. CH. 1.

I.

CL. 1. S. E. 1.

THE PUSS-MOTH. Though the *Caterpillar* of this Fly seems most fond of the *Willow*, it is often found on the *Abele*, or *Poplar* Tree. It has a Power of stretching out, and putting back its Head, at Pleasure; which it seems to do as it thinks itself more or less in Danger. It is armed with two Tubes at the End of the Tail, and two others within those, of a Blood red Colour, which it nimbly, if touched by any Thing, darts out: The Use of these is (as I apprehend) to prevent the *Ichneumon* Flies from settling on them; but, notwithstanding those Flies are often able to remain long enough to leave their Eggs behind: In such Case, the *Moth* is not produced, but, instead thereof, the *Ichneumon* pictured in the Plate. Such *Caterpillars* as the *Ichneumons* have not fixed their Eggs on, usually go into the *Chrysalis* State at the Beginning of *August*, on the Body of the Tree, and generally near the Ground, the *Caterpillar* gnawing the Wood into a Kind of Sawdust; which, mixing with a glutenous Matter, or Cement, that comes from its Mouth, makes a hard Case, in which the *Chrysalis* is formed, on the outside of the Tree; the *Moth* appears the *May* following. These *Moths* are seldom taken in the Fly-state; but may easily be obtained by breeding the *Caterpillars*, which may be found at the same Time, and in the same Places, where you take those that produce the *Willow* and *Poplar* Hawks.

Figure 2. in the same Plate, represents a *Moth* called the KITTEN. The *Caterpillar* feeds on *Black Thorn*, *Willow*, &c. becomes a *Chrysalis* at the Beginning of *August*, and the *Moth* is bred in *May*. The *Caterpillar* and Fly are rarely to be met with.

Cl. 1. S. E. 1.

Salix.

CL. 1. S. *E.* 3.

THE WILD ROSE-MOTH. The *Caterpillar* feeds on the *Sweet Briar* sometimes, but most commonly on the *Oak*; changes to a *Chrysalis* (within the Leaves of such Trees fastened together by a Spining) at the End of *September*, and the *Moth* is bred in the Middle of *May*. The *Caterpillar* and Fly are both scarce.

See *Albin*, P. 65. *Reaumur*, Pl. 22. Pag. 282. Vol. 2.



L. I. ch. 1.

Cl. 1. S. E. 3.

The Wild Rose, or Sweet Briar.

Rosa sylvestris; foliis odoratis. C. B. Pin.



C L A S S II.

CATERPILLARS *having little* HAIR.

S E C T . *a*.

Without Protuberances.

L. 1. CH. 1.

I.

CL. 2. S. *a*. 1

THE GOAT-MOTH. The most likely Means of obtaining the *Caterpillars* is, to separate the Bark of some old *Willow* from the Body of the Tree by a Chissel. Thus probably you may discover some of them: Put them into an earthen Jar, and feed them with *Willow* Wood, which Food they seem most fond of; but cover the Jar lest they creep away. Do not keep them in a Box, or any Thing made of Wood, lest they eat their Way through it and be lost. These *Caterpillars* are full fed about the Middle of *May*, at which Time they gnaw the Wood into a Kind of Sawdust, which, being mixed with a glutenous Matter from their Mouths, makes a Case wherein they change to the *Chrysalis*. They lye in that State for about six Weeks; after which the *Moths* appear, and may be found on the Bodies of *Willow* Trees. This *Chrysalis*, by the Help of strong, short Saw like Teeth round Part of each Joint, and pointing towards the Tail, is enabled to force its Fore Part through the Body of the Tree, (as is expressed in the Plate), near which you'll find the *Moth*, if you chance to look that Day it is bred. They are reported to be three Years proceeding from the Egg to the Fly; but this I have not myself experienced.

See *Lister* on *Godart*. P. 49. N. 39. *Albin*, Pl. 35. *Rosel*, Cl. 2. Tab. 18. *Reaumur*, Pl. 17. Pag. 342. Vol. 1.



L. I. ch. I.

Cl. 2. S. a. 1.

The Willow-tree.

Salix.

L. 1. CH. 1.

II.

CL. 2. S. a. 2.

THE EMPEROR-MOTH. The best Way to be furnished with this fine *Moth* is, to breed it from the *Caterpillar*, which may be found in its infant State during the Month of *May*; but it is most adviseable to seek for

them before the 20th. The *Caterpillars*, when newly hatch'd, are sociable, and keep together for about three Weeks, after which Time they separate and are difficult to be found. The most certain Food to take them on is the *Black Thorn*, and of that such as is not of above two or three Years Growth. They are usually found on Twigs about one or two Feet from the Ground, and (as I have observed) herd in Clusters together, when they are in their first, second and third Skins.—On the 12th of *May* 1745, I had about sixty *Caterpillars* hatched from such Sort of Eggs as are expressed round the *Black Thorn* in the Plate: On the 19th they appeared in their second Skin; on the 25th, in their third; on the 18th of *June*, in their fourth, and, on the 22d, in their fifth. The *Caterpillars* grew so much in this Skin in four Days, that they were three Times as big as when full fed in their fourth Skin. On *June* the 30th, they shifted into their sixth and last Skin; in which Skin (as well as in the former Skins) such as appear most Green are the female *Caterpillars*. On the 12th of *July* some of them began to spin Cases, in order for their Change into the *Chrysalis* State; and, on the 26th, the whole threescore had spun themselves up. Thus they remained throughout the Winter, and, from the 16th to the 30th of *April* following, I had forty of the *Moths* bred; but the Rest of the *Aurelias* did not produce their *Moths* till the Year following: So that about twenty of my Number were in the *Chrysalis* State near two Years; and yet produced *Moths* at last that were equally beautiful and large. The Female lays from 250 to 300 Eggs, in such Manner as is represented in the Plate, and these she hides in three or four different Places for the more certain Security of some of them.

The Case which contains the *Chrysalis* is well worthy Notice, being composed of a glutenous Matter fit to resist all Kinds of Weather; and, for its greater Security, that Part which I call the Mouth is doubly guarded: So that, if any Insect should chance to force the first Fence, it will there meet with a second Resistance, and that of so strong a Nature, from its particular make, that I am apt to believe few, if any, can ever enter; and thus the *Chrysalis* is preserved in the most secure and wonderful Manner, as will be better understood by attending to the Figure represented in the Plate. The *Moth* is very seldom taken in the Fly State.



L. 1. ch. 1.

Cl. 2. S. a. 2.

The Black Thorn.

Prunus Sylvestris. Ger.

L. 1. CH. 1.

III.

CL. 2. S. a. 3.

THIS Plate represents the *Male* and *Female* of the EMPEROR-MOTH, of which I have already given an Account.

See *Merian*, Vol. 1. Ch. 23. *Albin*, Pl. 25. *Rosel*, Cl. 2. Tab. 4, 5.
Reaumur, Pl. 49 and 50. Pag. 554. Vol. 1.



L. 1. ch. 1.

Cl. 2. S. a. 3.

The White Fruit-bramble.

Rubus vulgaris major, fructu albo. Raii. Syn.

L. 1. CH. 1.

IV.

CL. 2. S. a. 4.

THE PLUMB-TREE MOTH. The *Caterpillar* feeds in Gardens on various Fruit-trees: It changes to a *Chrysalis* about the End of *May*, within Leaves spun together, and lies in that State till the Middle of *June*, when the *Moth* appears.

See *Albin*, Pl. 36.



L. I. ch. I.

Cl. 2. S. a. 4.

The red Roman Nectarine.

Nucipersica Romana rubra.

S E C T . *b*.

CATERPILLARS, hairy, *with* Protuberances.

L. 1. CH. 1.

I.

CL. 2. S. *b*. 1.

THE WILLOW RED-UNDERWING MOTH. The *Caterpillar* feeds on *Willow*; changes to a *Chrysalis*, betwixt the Bark and Body of the *Willow* Tree, about the Middle of *June*. The *Moth* is bred in the Middle of *July*, and may be found at that Time on the Barks of *Willows*, and other Trees, as also against Walls, Houses, Barns, &c. If the Weather proves very hot, the *Moth* will fly in the Day-time. The *Caterpillar* is rarely found.

See *Albin*, P. 80. *Rosel*, Cl. 2. Tab. 15.



L. l. ch. 1.

Cl. 2. S. b. 1.

The weeping Willow.

Salix orientalis, flagellis deorsum pulchre pendentibus. Tourn. Cor.

C L A S S III.

C A T E R P I L L A R S *having much* H A I R.

S E C T . A.

Without Protuberances.

L. 1. CH. 1.

I.

CL. 3. S. A. 1.

THE GREAT TYGER MOTH. The *Caterpillars* are hatch'd from the Egg in *August*, and feed in their infant State on *Nettles*, *Lettice*, &c. till the End of *October*; when, if the Weather be cold and bad, they find themselves a Residence for the Winter in Holes and private Recesses, where they continue till the *March* or *April* following, as the Spring is more or less forward. At their coming out of their Holes they feed on the young *Nettles*, *Chickweed*, &c. that grows in dry Ditches and Gardens; and, when they are in their last Skin, which is usually at the End of *May*, they may easily be found in the Places already mentioned. At the Beginning of *June* they spin a Bag, and within that change to the *Chrysalis*, in which State they lye for one Month, after which the *Moths* come abroad and are very common. I have observed, that most of the *hairy Caterpillars* live throughout the Winter, after the Manner above mentioned.

See *Lister's Godart*. N. 99. *Moufet*. 15. P. 93. *Raii Hist. Insect*. P. 151. N. 3 and P. 152. N. 7. *Merian*, Vol. 1. Ch. 5. *Albin*. Pl. 20. *Rosel*, Cl. 2. Tab. 1. *Reaumur*, P. 534. Vol. 1.



L. 1. ch. 1.

Cl. 3. S. A. 1.

The Hatfield Plumb.

Prunus, fructu magno crasso subacido. Tourn.

L. 1. CH. 1.

II.

CL. 3. S. A. 2.

THE CREAM-SPOT TYGER MOTH. *Chickweed, Nettles, Lettice, &c.* are the Food of this *Caterpillar*, which is to be found on such Growths in dry Ditches, in Woods, &c. at the End of *April*. It lives all Winter as the last mentioned, and changes to a *Chrysalis*, within a *Spinning*,

at the Beginning of *May*. The *Moth* is bred about three Weeks after; flies in the Day-time, and is very common.

See *Merian*, Vol. 1. Ch. 6. *Albin*, Pl. 21. *Reaumur*, Pl. 31. Pag. 534. Vol. 1.



L. l. ch. 1.

Cl. 3. S. A. 2.

Chickweed.

Alfine media. C. B. 250.

L. 1. CH. 1.

III.

CL. 3. S. A. 3.

THE SCARLET TYGER MOTH. The *Caterpillar* lives all Winter, and is to be taken feeding on the *Hounds-Tongue*, *Nettles*, and *White Archangel*, at the End of *April*. If you look for them from *Charlton-Church*, in *Kent*, down to the Road that leads to *Woolwich*, there you will find them in Plenty. They make a Spinning in *May* amongst the dead Leaves on the Ground, and there change to the *Chrysalis*. The *Moths* come forth in *June* and fly by Day.

See *Merian*, Vol. 2. Ch. 8. *Albin*, Pl. 22.



L. I. ch. 1.

Cl. 3. S. A. 3.

The Hounds Tongue in Flower.

Cynoglossum majus, vulgare. C. B. 257.

L. 1. CH. 1.

IV.

CL. 3. S. A. 4.

THE BLACK ARCHES, a MOTH. The *Caterpillar* of this *Fly* remains in Egg all Winter, and hatching about the Middle of *April*, feeds on the Leaves of the *Oak*, *Apple* Trees, &c. A *Caterpillar* brought up by Mr. *Cabrier* came to its full Growth near the Middle of *June*; at which Time it fastened itself, in the Corner of the Box, by a few small Threads: The 14th it changed to a *Chrysalis*, and the *Moth* appeared the 4th of *July*. The *Caterpillar* was fed, in its infant State, with *Lettice*, there being no *Oak* Leaves at that Time. This *Moth* is to be found sticking on the Barks of the *Oak* Trees in *Richmond*, *Bushey*, and other Parks, in *July*; but is not common.

See *Merian*, Vol. 2. Ch. 22.



L. 1. ch. 1.

Cl. 3. S. A. 4.

The Apple Tree in Blossom.

Malus.

L. 1. CH. 1.

V.

CL. 3. S. A. 5.

THE GREAT ERMINE MOTH. Its *Caterpillar* feeds on most Kinds of Plants and Flowers; turns to a *Chrysalis*, within a Web on the

Ground, in *September*: The *Moth* is bred in the *May* following, and may be found sticking on the Barks of Trees, on Pales, Walls, &c.



L. L. ch. 1.

Cl. 3. S. A. 5.

The small Garden Sun-Flower.

Corona Solis, perennis & vulgaris. Vaill.

L. 1. CH. 1.

VI.

CL. 3. S. A. 6.

THE SPOTTED BUFF-MOTH. The *Caterpillars* of this *Moth* are very mischievous in Gardens, feeding on most Kind of Herbage; but may easily be shaken off the Plants and killed. They change to the *Chrysalis* within a Web on the Ground, and sometimes within the Ground, in *September*, and the *Moths* came out in *May*. Their Eggs are of a light green Colour; and the *Caterpillars*, when first hatch'd, which is about the Middle of *June*, are white and hairy. The *Moth* is frequently taken sticking on Houses, Walls, Pales, &c.



L. L. ch. 1.

Cl. 3. S. A. 6.

Arsmart.

L. 1. CH. 1.

VII.

CL. 3. S. A. 7.

THE GIPSEY MOTH. The *Caterpillars* are very destructive to all Fruit-trees, on which they are found in Swarms envelop'd with a Web like that of the Spider. In the Day-time they disperse themselves over the whole Tree, but at Night come together again, and retire to their Web. They put on the *Chrysalis*, form within a Spinning, at the Beginning of *July*, and the *Moth* is bred about fourteen Days after. Their Eggs are covered with a yellowish Down, whereby they are preserved from Cold throughout the Winter. This *Moth* is very common in *Germany*, and was produced from a Nest of Eggs that were sent to Mr. *Peter Collinson*, who gave them to *Charles Lockyer*, Esq; He bred *Moths* from them as above mentioned; and, having turned Numbers of them wild (as I have been informed) about *Ealing* near *Brentford*, in *Middlesex*, they are to be found there, but not any where else that I have yet heard of.

See *Rosel*, Cl. 2. Tab. 3. *Merian*, Vol. 1. Ch. 18.



L. 1. ch. 1.

Cl. 3. S. A. 7.

The Crab-tree.

Malus sylvestris. Ger. 1276.

L. 1. CH. 1.

VIII.

CL. 3. S. A. 8.

THE BUFF-TIP MOTH. The *Caterpillar* feeds on the *Willow* and *Oak*, and may be easily found amongst the *Willows* growing about *Chelsea*, and in such like Places. They may be taken full fed towards the End of *August*; at which Time Numbers of them go into the Earth, and there

Change to the *Chrysalis*, and the *Moths* appear in the following May. It is not a very common Thing to take this *Moth* flying.

See *Lister's Godart*, N. 95. *Raii Hist. Insect.* P. 162. N. 14. *Albin*, Pl. 23. *Rosel*, Cl. 2. Tab. 14.



L. I. ch. 1.

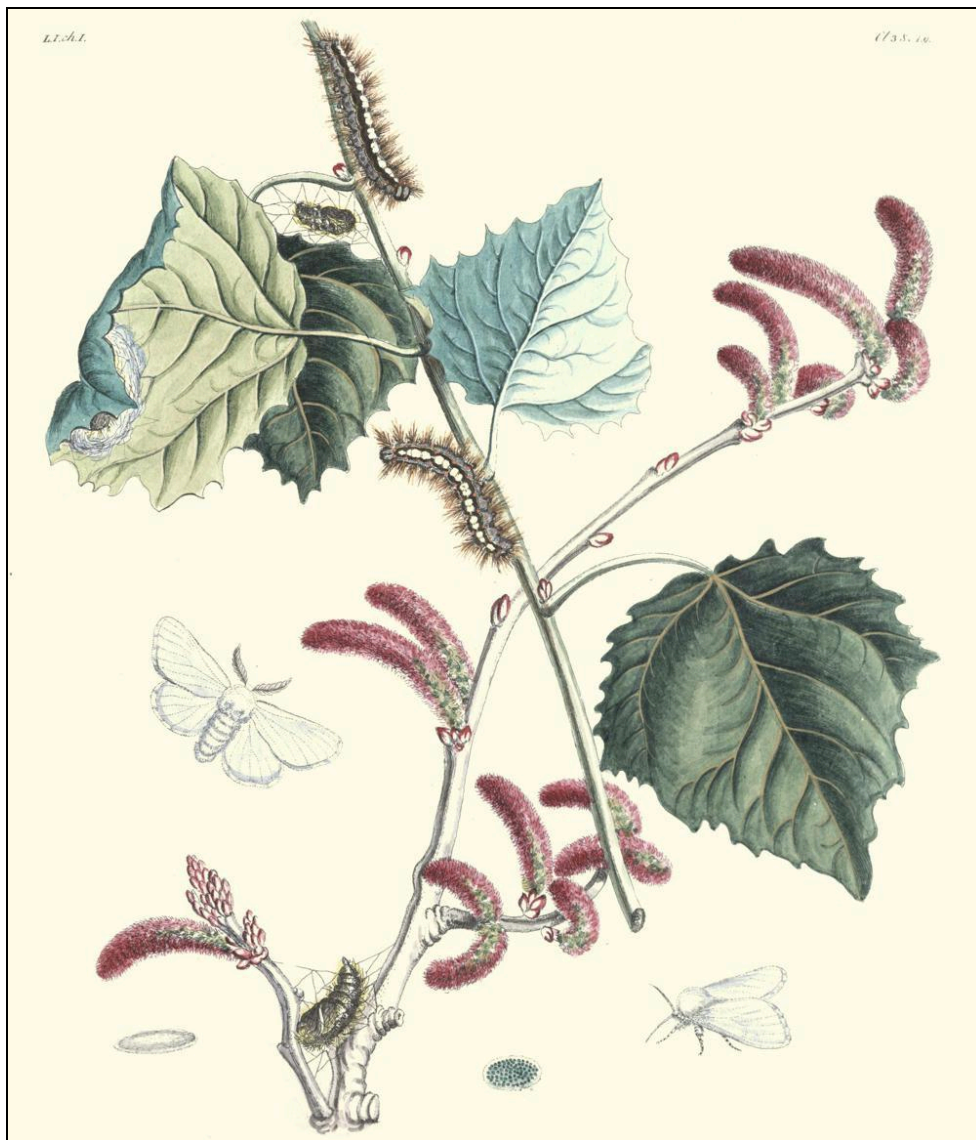
Cl. 3. S. A. 8.

The Yellow Dwarf-Willow.

Salix folio longo subluteo, non auriculata, viminibus luteis. Raii. Syn.

THE WHITE-SATIN MOTH. Its *Caterpillar* feeds on the *White Poplar*, as also on *Willow*; changes to a *hairy Chrysalis*, within the Leaves spun together, in *June*; lies in that State about twenty Days, and then the *Moth* flies abroad. You may take these Flies by shaking the Boughs of the Trees they feed on. Their Eggs, when first laid, are of a pale green Colour, which they cover with a Substance looking like Leaf-silver. They continue about one Month in the Egg-state, and the *Caterpillars* live all Winter.

See *Lister* on *Godart*. P. 84. N. 87. *Rosel*, Cl. 2, Tab. 9. *Reaumur*, Pl. 34. Pag. 534. Vol. 1.



L. 1. ch. 1.

Cl. 3. S. A. 9.

The White Poplar-Tree.

Populus alba majoribus foliis. C. B.

L. 1. CH. 1.

X.

CL. 3. S. A. 10.

THE LACKEY-MOTH. The *Caterpillars* are very common, and feed on most Kind of Herbage, especially *Black Thorn*, *White Thorn*, &c. about the Middle of *June* they change to the *Chrysalis* within a Case, which they spin, and cover with a fine Dust of the Colour of Brimstone. In this State they remain for about one Month; then the *Moth* appears, and lays her Eggs, in a spiral Order, round the young Twigs of such Trees as are proper Food for the infant *Caterpillars*. They hatch in the Spring, and keep together in a Web while young. As these *Caterpillars* are destructive in Gardens to Fruit-trees, &c. its adviseable to destroy them.

See *Godartius*, P. 1. Tab. 10. *Albin*, Pl. 19. *Rosel*, Cl. 2. Tab. 6. *Reaumur*, Vol. 2. Pl. 4. Pag. 120.



L. 1. ch. 1.

Cl. 3. S. A. 10.

Figure 1. Tway Blade. 2. The Crane's Bill. 3. The Yellow-Vetchling.

1. *Bifolium*, vulg. C. B. 2. *Geranium Robertianum*. 3. *Lathyrus luteus sylvestris dumetorum*. 1. B. II. 304.

L. 1. CH. 1.

XI.

CL. 3. S. A. 11.

THE GREAT EGGER-MOTH. The *Caterpillars* of this *Moth* are to be found, full grown, towards the End of *May*, feeding on *Black Thorn* and *White Thorn*. At the Beginning of *June*, they spin a brown Case, in the Shape of an Egg, in which the *Chrysalis* is formed: They remain in that State for about one Month; then the *Moth* is produced. The Males fly swiftly in the Day-time, from 12 o’Clock to 5 in the Afternoon. The Eggs are of an oval Shape, and usually hatch at the latter End of *July*. When the *Caterpillars* are young, you may feed them with *Black Thorn*, or *White Thorn*, and, during the Winter, with *Bramble*, *Holly*, *Privet*, &c. When you breed any female *Egger-Moths*, take one of them and put it in a Box (covered with a Piece of Crape, to prevent its getting away), carry the Box with you to the Sides of Woods and green Lanes; and, if the Day be fair, and the Sun shines bright, put it upon the Ground: And, if there is a Breed of these *Moths* near, the Males will come and endeavour to get at the Female. By this means I have taken with my Net near twenty in an Hour. This Method of taking Flies is called *Simbling*; and many other Kinds of *Moths* are to be caught the same Way.

See *Godart*. P. 1. Tab. 7. *Lister’s Godart*. N. 88. *Merian*, Vol. 1. Ch. 10. *Moufet*, P. 92 N. 9. *Raii’s Hist. Insect*. P. 142. N. 2. *Albin*, Pl. 18. *Rosel*, Cl. 2. Tab. 35. a. *Reamur*, Pl. 35. Pag. 534. Vol. 1.



L. I. ch. I.

Cl. 3. S. A. 11.

The White Thorn in Flower.

Mespilus sylvestris, seu Oxyacantha.

L. 1. CH. 1.

XII.

CL. 3. S. A. 12.

THE GRASS EGGER-MOTH. I took the *Caterpillars* of this *Moth* feeding on *Clover* amongst *Grass*. They make a *Spinning* and *Case*, on the *Surface* of the *Earth*, about the *Middle* of *June*, and therein turn to a

Chrysalis. The *Moth* flies in the Middle of *August*; but is not commonly met with.

See *Rosel*, Cl. 2. Tab. 35. b.



L. I. ch. I.

Cl. 3. S. A. 12.

The double flowering Cherry, and Grass.

Cerasus hortensis, flore roseo. C. B. Pin. 450.

L. 1. Ch. 1.

XIII.

CL. 3. S. A. 13.

THE DECEMBER MOTH. Mr. *Dandridge* took some of its *Caterpillars*, at the Beginning of *June*, on the *Hasle*; but, observing they eat but little, gave them fresh Boughs that were wet with Rain, on which they fed greedily. The next Day he gave them Water and Honey mixt together, of which they drank a good Quantity, and fell to eating again very heartily. In this Manner he fed them every Day, and sometimes twice a Day, till the Middle of the same Month, when they went into the Earth and changed into *Chrysalis*. The *Moths* came out at the End of *December*. I took this Year (1748) two of these *Caterpillars* on the Leaves of an *Oak*, wherewith I fed them till the Middle of *June*, when they spun a dirty coloured Case which contained the *Chrysalis*; and the *Moths* were bred on the 24th *October*.

See *Albin*, P. 85. *Rosel*, Cl. 2. Tab. 60.



L. I. ch. I.

Cl. 3. S. A. 13.

The Nut-tree.

Corylus, Sativa, fructu anguloso. C. B. 418.

L. 1. Ch. 1.

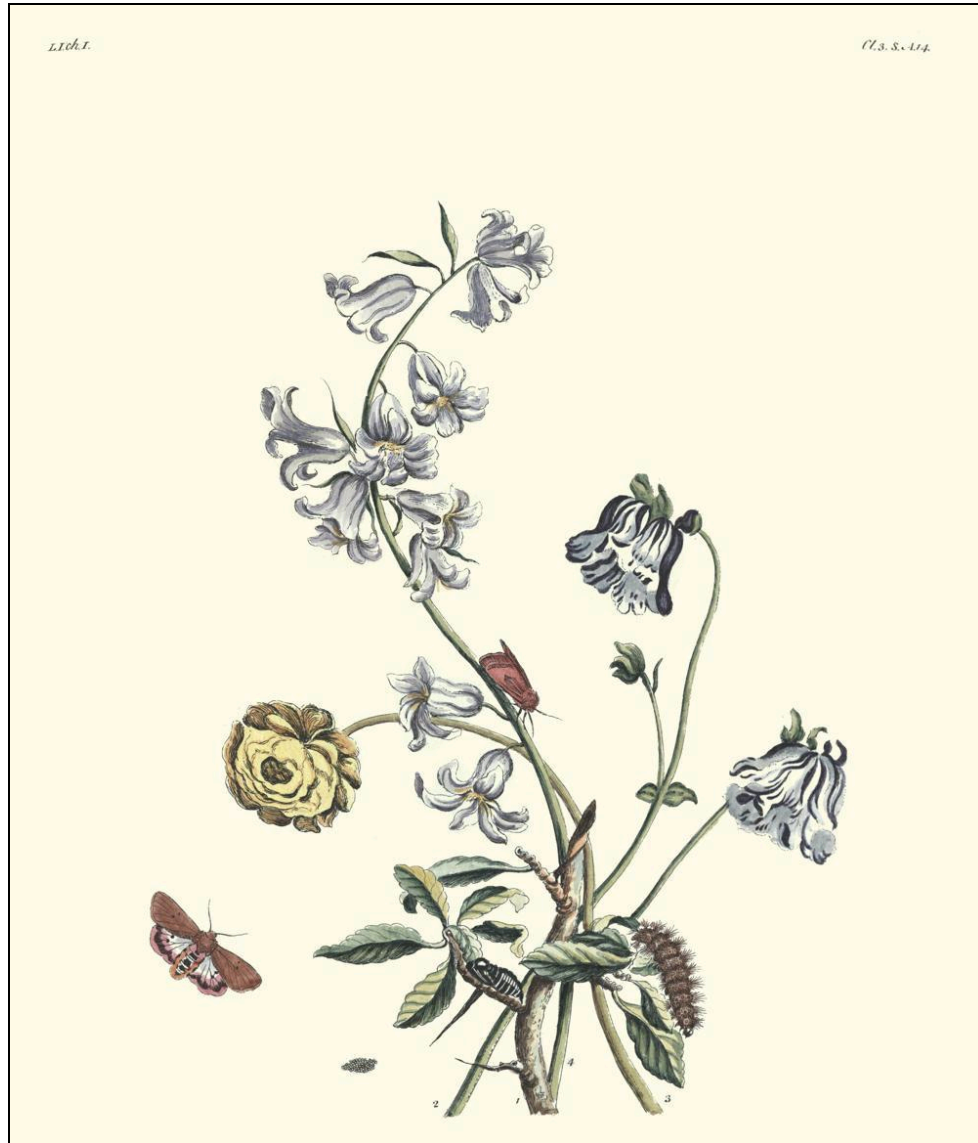
XIV.

CL. 3. S. A. 14.

THE SPOTTED RED and WHITE UNDERWING MOTH. I found and fed the *Caterpillar* on *Black Thorn*: It made a Spinning, and changed to *Chrysalis* in the Middle of *April*, and the *Moth*

came abroad about the Middle of May. The *Caterpillar* lives all Winter, and the *Moth* is rarely to be met with.

See *Rosel*, Cl. 2. Tab. 43.



L. I. ch. I.

Cl. 3. S. A. 14.

Figure 1. The Black Thorn. 2. The Columbine. 3. The Ranunculus. 4. The White Bells.

1. *Prunus sylvestris*. C. B. 2. *Aquilegia vulgaris simplex*. C. B. 3. *Ranunculus*. 4. *Hyacinthus, flore albo*. C. B.

THE WOOD TYGER MOTH. This *Moth* I first discovered in *Cain Wood*. The *Caterpillar* lives all Winter, feeds on *Chickweed*, *Lettice*, the *Lesser Plantain*, &c. and is full grown about the Middle of *April*, when it changes to the *Chrysalis* within a Spinning: The *Moth* is bred in the Middle of *May*, flies by Day, and may be taken in *Cain* or *Tottenham Woods*; an Afternoon is the best Time.



L. 1. ch. 1.

Cl. 3. S. A. 15.

The Vredenrick Hyacinth, and Lesser Plantain.

Hyacinthus Belgicus Vredenrick dictus. Plantago minor. I. B.

THE MOTHER OF PEARL MOTH. *Nettles* are the Food of the *Caterpillar*, within the Leaves of which (folded together) it lives, and feeds till the Beginning of *June*; when it puts on the *Chrysalis* Form wrapt up within a Leaf, whose Edges are fastened by a Spinning: Fourteen Days after the *Moth* appears, and may be taken, very commonly, in an Evening, amongst *Nettles*.

See *Albin*, P. 73. *Rosel*, Cl. 4. Tab. 4.



The Cherry-Plumb.

Prunus, fructu majori, rotundo, rubro. Tourn.

L. 1. CH. 1.

XVII.

CL. 3. S. A. 17.

THE YELLOW JULY OAK-MOTH. The Food of the *Caterpillar* is the Leaves of the *Oak* Tree, on which it feeds till the Middle of *June*; then it changes to a *Chrysalis* in some Chink or other that it meets with in the Bark of the said Tree, over which Chink or Hole the *Caterpillar* spins a strong Web, which prevents any Bird or Insect from getting at it in its *Chrysalis* State. The *Moth* is bred in the Middle of *July*, and may, now and then, be found on the Barks of the *Oak* Trees in *Richmond* and other Parks, about that Time; but it is not a common Fly.

See *Rosel*, Cl. 2. Tab. 17.



L. 1. ch. 1.

Cl. 3. S. A. 17.

The May-cherry.

Cerasus, majalis, fructu duro subdulci. Tourn.

L. 1. CH. 1.

XVIII.

CL. 3. S. A. 18.

THE SMALL EGGER-MOTH. *Black and White Thorn* are the Food of the *Caterpillars*, which are always found in Colonies; the whole Nest of them feeding together till they are ready to change. When they are

first hatch'd from the Egg, they spin a Web to shelter themselves from the Weather; and, as they grow, enlarge it with several Divisions. They seldom go from the Plant they are first hatched on, till they have eaten it quite bare and want Food. They ordinarily spin a Thread from their main Web to guide them Home again; this is very particular, and seems necessary to them. About the Beginning of *July* you may find them full fed; then they spin a Web, within which a Case is formed, shaped like an Egg, which contains the *Chrysalis*: In this State they remain all Winter, and the *Moths* are bred in *March*. It is adviseable, if you feed the *Caterpillars*, to take the whole Nest with the Web. I have seldom taken the *Moths*.

See *Rosel*, Cl. 2. Tab. 62. *Albin*, P. 19.



L. 1. ch. 1.

Cl. 3. S. A. 18.

The Almond Tree, Blossom and Fruit.

Amygdalus sativa fructu majore. C. B. Pin. 441.

L. 1. Ch. 1.

XIX.

CL. 3. S. A. 19.

THE FOX COLOURED MOTH. The *Caterpillar* is hatch'd in *July*, lives all *Winter*, and feeds on *Grass*, *Brambles*, &c. I have seen them in *September* as large as they are in the *Middle of April*, which is the Time

they go to *Chrysalis*, within a Spinning or Web. The *Moth* is bred in the Middle of *May*, and the Cock flies swiftly in an Afternoon; but may be taken in a Net, by the Sides of Woods and green Lanes.

See *Albin*, P. 81.



L. I. ch. I.

Cl. 3. S. A. 19.

The Blackberry Fruit.

Rubus, fructu nigro.

SECT. 2 A.

CATERPILLARS *having Rings of different Colours surrounding them.*

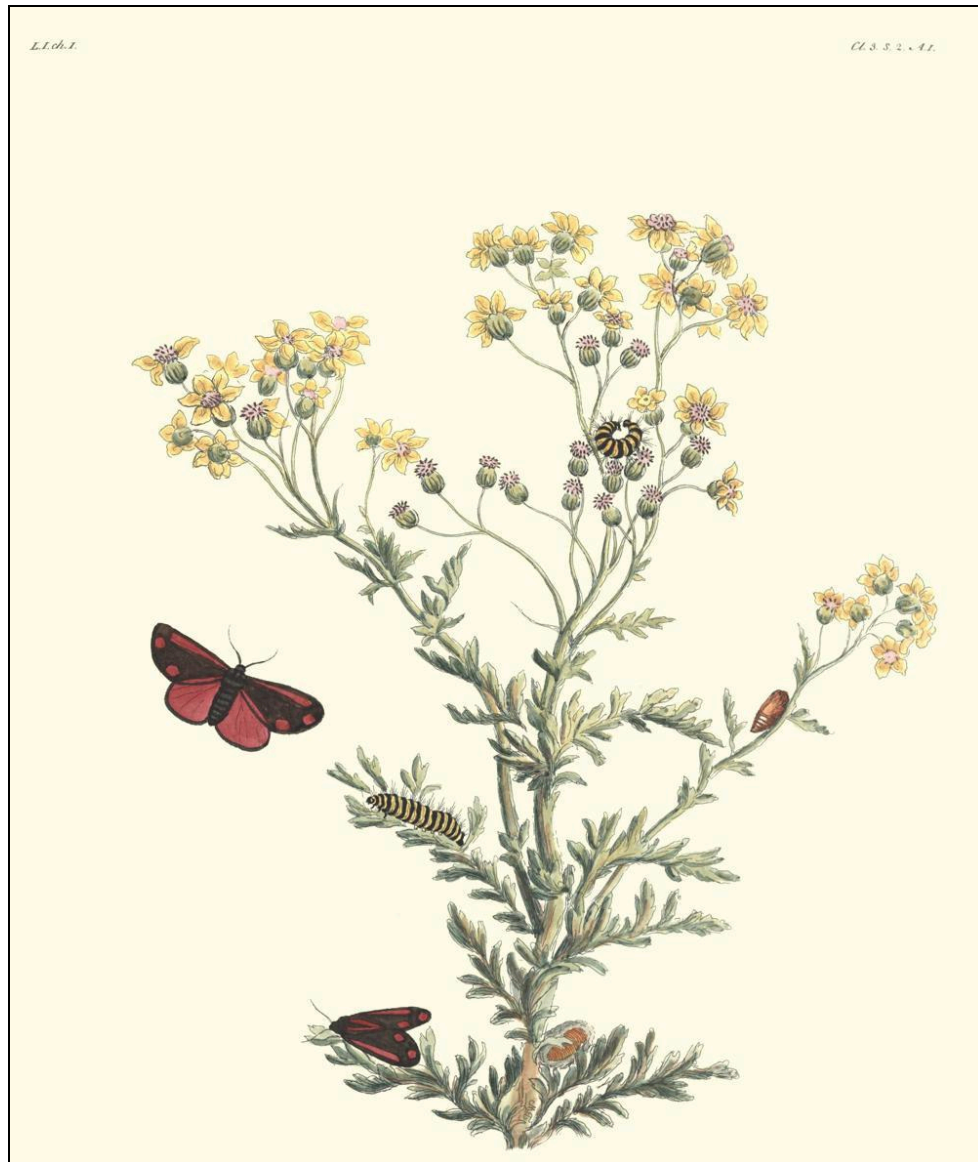
L. 1. CH. 1.

I.

CL. 3. S. 2 A. 1.

THE CINNABAR-MOTH. The *Caterpillars* of this Fly are sociable and feed together in Numbers on the *Ragwort*; in *July* they are full grown, and then some go into the Earth, and others spin on the Surface of the Ground, in order to their Change into the *Chrysalis* State. The *Moths* are produced in the *May* following, and are common in Fields and Gardens.

See *Lister* on *Godart*. P. 61. N. 54. *Albin*, Pl. 34. *Rosel*, Cl. 2. Tab. 48. *Reaumur*, Pl. 16. Pag. 342. Vol. 1.



L. I. ch. I.

Cl. 3. S. 2. A. 1.

The Ragwort, in Flower.

Jacobea, vulgaris. J. B. H. 1057.

SECT. 3 A.

CATERPILLARS *having* indented Markings.

L. 1. CH. 1.

I.

CL. 3. S. 3 A. 1.

THE BRAMBLE MOTH. Its *Caterpillar* is to be found feeding on the *Bramble*, on the *Oak*, &c. at the Beginning of *September*; and, towards the Middle of the same Month, it changes to a *Chrysalis*, mixing its Hairs with the Bag it spins: The *Moth* comes forth the *May* following, and is not very common.



L. 1. ch. 1.

Cl. 3. S. 3. A. 1.

The Orange Apricock.

Malus Armeniaca, fructu aurantio.

SECT. *b*.

CATERPILLARS *having* hairy, or fleshy Protuberances, *on the* Head, Back, or Tail.

L. 1. CH. 1.

I.

CL. 3. S. *b*. 1.

THE LAPPIT-MOTH. The Eggs of this *Moth* are laid about the Middle of *July*, and remain in that State fourteen Days. I fed the *Caterpillars*, when first hatch'd, with *Black Thorn*, and continued giving them the same Food till *October*; at which Time they were in their fourth Skin, and appeared as represented at Letter *a*; They there stretched themselves at full Length on the Twigs, and remained so till the *April* following, without eating any thing that I could perceive, notwithstanding at different Times I put in fresh green Food to try them. About the End of *May* the *Caterpillars* are full grown, in which Condition *b* represents the Female; they then make a Spinning, within which the *Chrysalis* is formed, and remains in that State for one Month, when the *Moth* is bred; though it is rarely met with in the Fly-state. The *Caterpillars* are to be found, at the End of *May*, sticking close to the Boughs of the *Black Thorn*, *Bramble*, &c. You must look for them about a Foot or two from the Ground; but, as their Colour nearly resembles the Bark of their Food, it is very difficult to see them.

See *Albin*, Pl. 16. *Rosel*, Cl. 2. Tab. 41. *Merian*, Vol. 1. Ch. 17. *Reaumur*, Pl. 23. Pag. 322. Vol. 2.



L. 1. ch. 1.

Cl. 3. S. b. 1.

The Privet leaved Phylerea.

Phyllyrea, Ligustri folio. C. B.

L. 1. Ch. 1.

II.

CL. 3. S. b. 2.

THE DRINKER-MOTH. Its Eggs are usually laid about the Middle of *July*; in about fourteen Days the *Caterpillars* come out: They feed in the Winter-season on long *Grass*, &c. and change to *Chrysalis* at the End of

May (within such a Case as is expressed in the Plate) at which Time they are easily found upon the Grass, by the Sides of Fields and Hedges. The *Moths* are bred at the End of *June*, and may be taken flying in an Evening, in green Lanes and by the Sides of Woods.

See *Lister's Godart.* N. 82. *Albin*, Pl. 17. *Raii Hist. Insect.* P. 142. N. 3. *Rosel*, Cl. 2. Tab. 2.



L. I. ch. 1.

Cl. 3. S. b. 2.

*Figure 1. The Yellow Vetchling. 2. Bistort. 3. The
White Darnel Grass. 4. The Wood-Pease.*

1. *Lathyrus sylvestris, dumetorum, flore luteo.* C. B. 2. *Bistorta major.* I.
B. 3. *Gramen loliaceum spica longiori.* C. B. 4. *Astragalus sylvaticus.* Ger.
Emac.

L. 1. CH. 1.

III.

CL. 3. S. b. 3.

THE YELLOW-TAIL MOTH. The *Caterpillar* feeds on *Elder*,
Oak, *Black Thorn*, &c. upon all which Growths you may find them at the
End of *May*. It changes to a *Chrysalis*, within a Spinning, at the Beginning
of *June*, and the *Moth* comes forth at the End of the same Month. They may
be found sticking against the Barks of the Trees in Parks, and other Places,
and also flying in the Evening. They lay Eggs of a pale Colour, and cover
them with a woolly Substance. The *Caterpillars* are hatch'd in *August*, and
will feed on *Black Thorn* till the End of *October*; at which Time they spin
themselves little Cases, and remain in them till the Spring following.

See *Rosel*, Cl. 2. Tab. 21. *Reaumur*, Pl. 16. Pag. 342. Vol. 1.



L. 1. ch. 1.

Cl. 3. S. b. 3.

The Alder-Tree.

Alnus, rotundifolia, glutinosa, viridis. C. B.

L. 1. CH. 1.

IV.

CL. 3. S. b. 4.

THE DAGGER MOTH. The *Caterpillar* feeds on Rose Trees, Fruit-trees, and Shrubs; becomes a *Chrysalis* within the Earth in *October*, and in *May* following the *Moth* appears, and may be found sticking against Pales,

Walls, Trees, &c. Their Eggs are very small, and of a fine green Colour, and the *Caterpillars* are common in Gardens and other Places. It is observed, that the Protuberance on the Back of the *Caterpillar* is of a fleshy Substance, with Hair growing out of it.

See *Albin*, P. 86. *Rosel*, Cl. 2. Tab. 7. *Reaumur*, Pl. 42. Pag. 602.



L. I. ch. I.

Cl. 3. S. b. 4.

The most double Sweet Briar Rose.

Rosa sylvestris, sive Eglantheria, flore pleno.

THE WILD PINE-TREE LAPPIT-MOTH. Mr. *Rosel* informs us, That the *Caterpillars* of this curious Fly feed on the *Pinus sylvestris*, or WILD PINE: That they are full fed in the Month of *June*, at which Time they change to *Chrysalis*, within such a Spinning as is represented in the Plate; and, after lying in that State three Weeks the *Moths* appear.

About the Middle of *September*, 1748, I took one of these *Caterpillars* upon a *White Thorn* Bush, near *Richmond* Park; which is still alive this 20th Day of *January* 1748-9, though it has eat nothing, that I know of, since it has been in my Custody. It remains stretched out, in the Manner of the *Lappit-Caterpillar* represented at the Letter *a* of the first Plate of this Section; and I do not think it will begin to feed until there are Buds of *Black* or *White Thorn* to give it. This is the only *Caterpillar* of the Kind that I have known taken in *England*.—For the Drawings exhibited in my Plate, I am obliged to Mr. *Rosel*.

See *Rosel*, Cl. 2. Tab. 59.



L. I. ch. I.

Cl. 3. s. b. 5.

The Wild Pine-Tree.

Pinus sylvestris. C. B. Pin.



S E C T . C .

CATERPILLARS, *having many Tufts of Hair, producing* TUSSOCK-MOTHS.

L. 1. CH. 1.

I.

CL. 3. S. c. 1.

THE BLACK TUSSOCK-MOTH. This *Moth* is bred about the Middle of *June*; at which Time it lays Eggs of a light grey Colour, inclosed within a woolly Substance; and thus they continue for about one Month. When the infant *Caterpillars* appear, they feed on *Black Thorn, White Thorn, Bramble, &c.* They live all the Winter in the *Caterpillar* State, and change to a *hairy Chrysalis*, within a Spinning, about the Middle of *May*. The *Caterpillars* may be found, full fed, on *Black Thorn* that grows by the Sides of Woods and Hedges; as for the *Moth*, it is not commonly taken in the Fly-state.

See *Albin*, Pl. 26. *Rosel*, Cl. 2. Tab. 37.



L. I. ch. I.

Cl. 3. S. c. 1.

The White double Garden-Rose.

Rosa Alba hortensis, flore pleno. I. B. 1144.

L. 1. CH. 1.

II.

CL. 3. S. c. 2.

THE YELLOW TUSSOCK-MOTH. Its *Caterpillar* feeds on the *Red Archangel*, *Oak*, *Hasle*, &c. towards the End of *September*. It makes a *Spinning*, within which a *hairy Chrysalis* is formed, where it lives till the

Middle of *May* following, and then the *Moth* takes its flight; but it is not a very common Fly.

See *Merian*, Vol. 1. Ch. 47. *Albin*, Pl. 26. *Rosel*, Cl. 2. Tab. 38. *Reaumur*, Pl. 33. Pag. 534. Vol. 1.



L. I. Ch. I.

Cl. 3. S. c. 2.

The Red Archangel.

Lamium rubrum. Ger. 568.

THE RED-SPOT TUSSOCK-MOTH. The *Caterpillars* are very common, feeding on most *Elm* and *Lime* Trees about Town. They make a Spinning under the Copings of Walls, Pales, &c. at the End of *May*; and the *Moth* is bred at the End of *June* and in *July*. The Females have no Wings; but, if you carry one or two of them in a Box (where there is a Breed of this *Moth*) the Males will follow you in order to get at them: So that, if you put the Box on the Ground, you may easily take the Cocks with your Net. Several other *Moths* may be taken by the like Means.

See *Albin*, P. 89. *Reaumur*, Pl. 19. Pag. 342. Vol. 1.



L. 1. ch. 1.

CL. 3. S. c. 3.

The smooth Leaved Elm.

Ulmus folio Glabro. Ger. Em. 1481.

L. 1. Ch. 1.

IV.

CL. 3. S. c. 4.

THE ORANGE TUSSOCK-MOTH. The Food of its *Caterpillar* is *Hasle*, *St. John's Wort*, *Oak*, &c. About the Middle of *May* it made a

Spinning, and changed to a *Chrysalis*, and the *Moth* came forth three Weeks after. It is very scarce, and the Female has no Wings. I took one of these *Caterpillars*, feeding on *Oak Leaves*, on the 25th of *September* 1748; two Days after, it spun up in order to its *Chrysalis* State, and, on the 8th of *October*, the *Moth* was bred: From whence I conclude, this *Moth* breeds twice in a Year.

See *Rosel*, Cl. 2. Tab. 40. *Albin*, P. 90.



L. I. ch. I.

Cl. 3. S. c. 4.

The elegant upright St. John's Wort.

Hypericum pulchrum Tragi. I. B. III. 383.

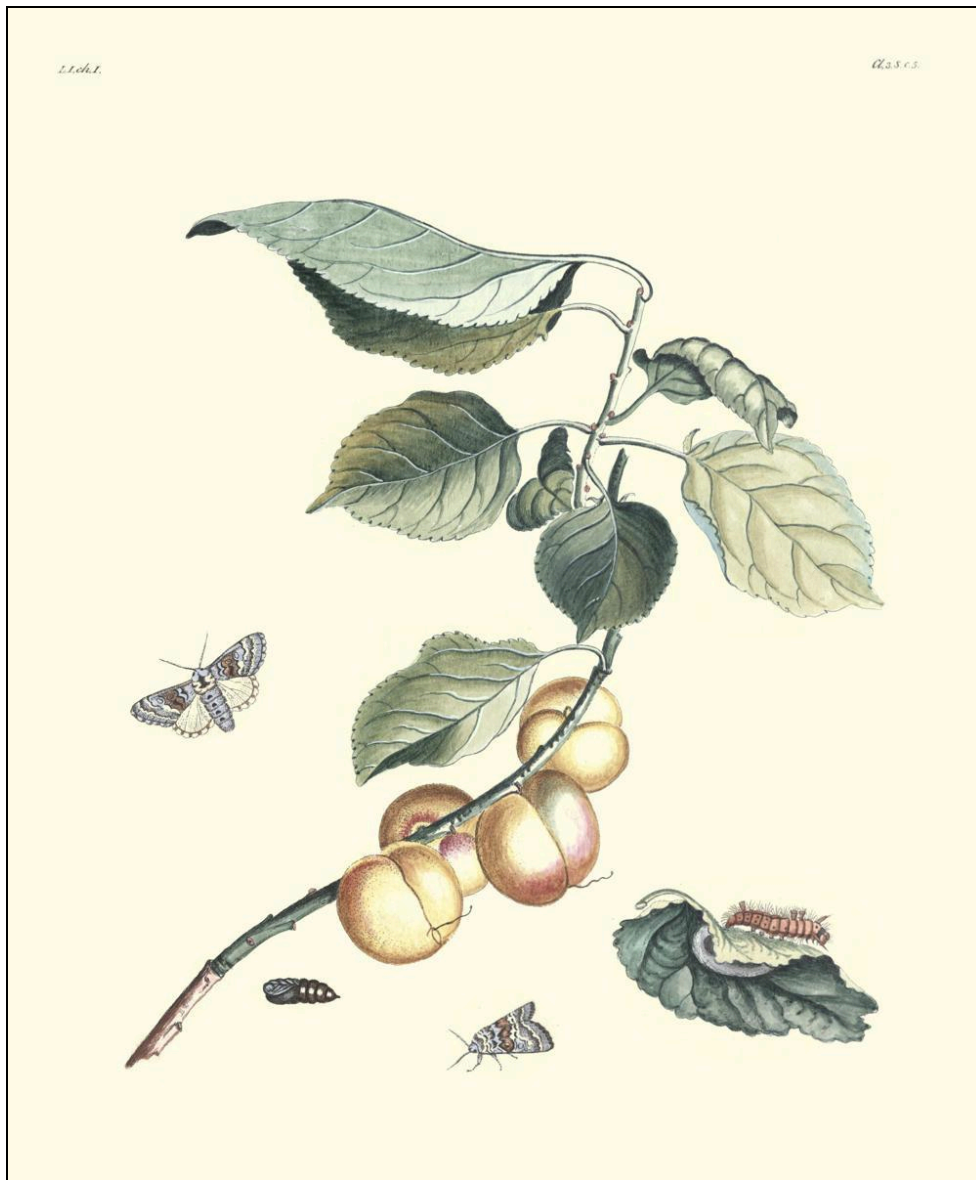
L. 1. Ch. 1.

V.

CL. 3. S. c. 5.

THE NUT-TREE TUSSOCK-MOTH. The *Caterpillar* of this very scarce Fly was fed upon the *Hasle*. About the Middle of *September* it spun up, and changed into a *Chrysalis*, and the *Moth* was bred at the Beginning of *April*.

See *Albin*, P. 90. *Rosel*, Cl. 2. Tab. 53.



L. 1. ch. 1.

Cl. 3. S. c. 5.

The Masculine Apricock.

Armeniaca, fructu minore præcocior. Miller.

L. 1. CH. 1.

VI.

CL. 3. S. C. 6.

THE SYCAMORE TUSSOCK-MOTH. You may find the *Caterpillars* on *Sycamore* Trees, and, towards the End of *August*, they are usually full fed; at which Time they spin themselves up, intermixing their Hairs with the Web, and change into the *Chrysalis* State. The *Moths* are bred in *May* and *June*, are not uncommon, and may be found on the Bodies of the Trees in *St. James's Park*, on Pales, Walls, &c.

See *Albin*, P. 83. *Reaumur*, Pl. 34. Pag. 534. Vol. 1.



L. I. ch. I.

Cl. 3. S. c. 6.

The Sycamore Tree.

Acer majus, multis falso Platanus. I. B.



C H A P. II.

CATERPILLARS that perform their progressive Motion after a different Manner, viz. by drawing up the Tail towards the Head, whereby their Bodies become bent into the Form of a *half Loop*, on which Account they are called HALF-LOOPERS.

C L A S S I.

CATERPILLARS *having* eight Feet, two Holders, *and* six Claws *or* Hooks, *and having* Protuberances.

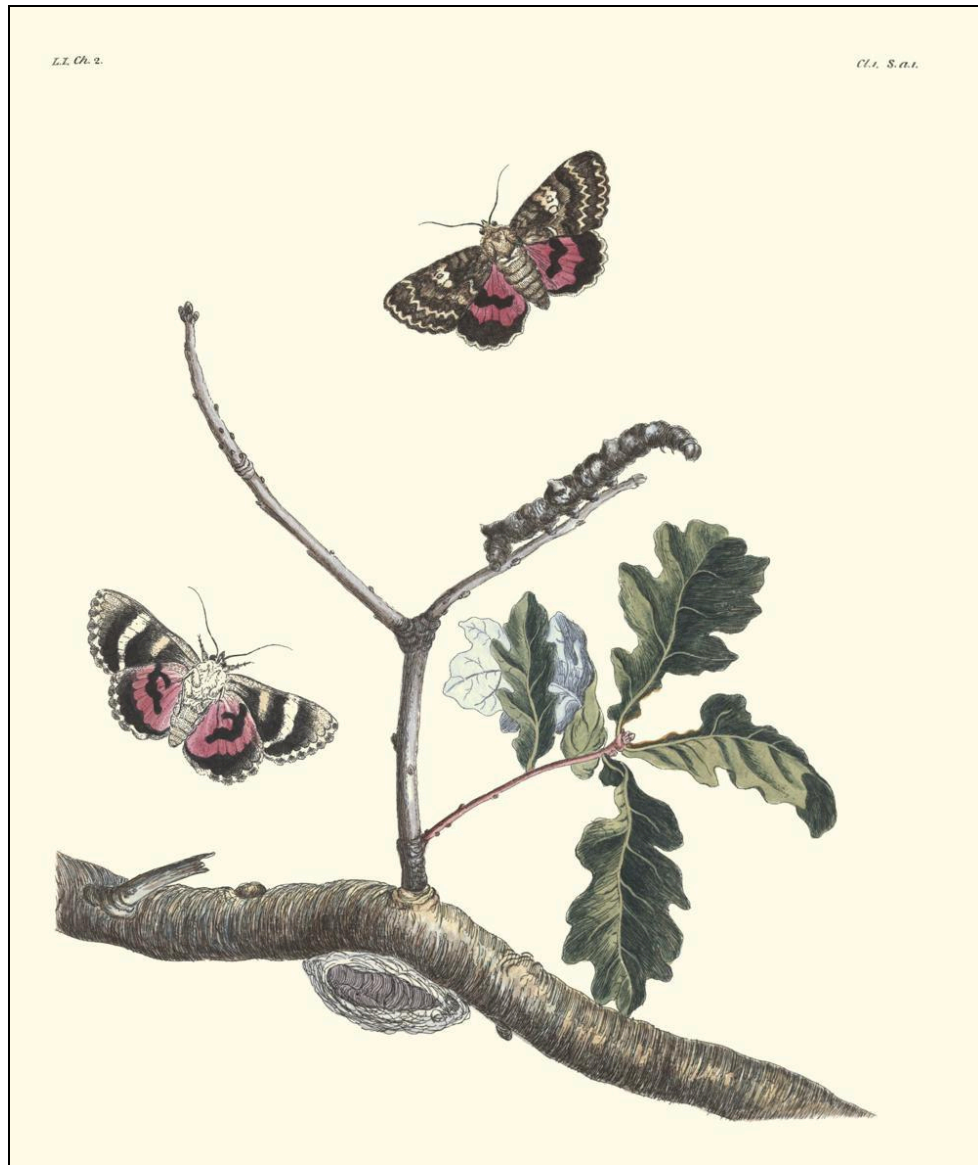
S E C T. a.

L. 1. CH. 2.

I.

CL. 1. S. a. 1.

THE CRIMSON UNDERWING MOTH. The *Caterpillar* of this *Moth* feeds on *Oak-leaves*, and becomes a *Chrysalis*, within some of the same Leaves spun together, at the Beginning of *June*. The *Moth* appears at the Beginning of *July*, and is extremely scarce, as is also the *Caterpillar*.



L. I. ch. 2.

Cl. 1. S. a. 1.

The Oak-Tree.

Quercus Latifolia. C. B.

Class II.

CATERPILLARS *which are hairy, having two Holders, four Feet, and six Hooks or Claws.*

SECT. *a.*

L. 1. CH. 2.

I.

CL. 2. S. *a.* 1.

THE SILVER Y MOTH. The *Caterpillar* feeds on *Sage, Clover, Burdock, &c.* changes to a *Chrysalis*, within a slight *Spinning*, in *August*, and the *Moth* is bred in *September*; is very common in most *Fields* and *Gardens*, and flies in the *Day* as well as at *Night*. There is an early *Breed* of this *Moth* in *May*.

See *Rosel*, Cl. 2. Tab. 5. *Albin*, P. 79. *Lister* on *Godart*. P. 20. N. 14. *Reaumur*, Pl. 19. Pag. 342. Vol. 1. Pl. 26. P. 348. Vol. 2.



L. I. Ch. 2.

Cl. 2. S. a. 1.

Common Sage in Flower.

Salvia hortensis. I. B.



C H A P. III.

THE *Caterpillars* of this Chapter have two Feet and two Holders, with six Hooks or Claws; but have no Feet in the Middle Part of their Bodies like those already described. Their progressive Motion is performed by fixing the Head-part, and drawing up the Tail to it; so that each Step, forming a Loop, they are called LOOPERS. They are provided with strong and sharp Claws in their Holders, by which they can fasten themselves in such manner as to support the Rest of their Bodies, either perpendicularly or horizontally, or in any other Posture they please, without any regard to the Center of Gravity in their respective Bodies. They are able likewise, out of their own *Viscera* or Bowels, to furnish themselves with Lines, whereby they can let themselves down from the Branches of Trees, or other Heights, as they find Occasion.

C L A S S I.

Smooth or naked Loopers.

S E C T. a.

Having no Protuberances.

L. 1. CH. 3.

I.

CL. 1. S. a. 1.

THE BRINDLED BEAUTY MOTH. The *Caterpillar* will feed on most Fruit-trees, as also on the *Willow*, &c. Its *Chrysalis* is formed within the Earth towards the End of *June*, and the *Moth* comes forth in *April*. I have taken Plenty of them sticking on the Barks of the *Willow* Trees about the Water-works at *Chelsea*.

See *Merian*, Vol. 1. P. 75. Chap. 37. *Albin*, Pl. 39.



L. I. ch. III.

Cl. 1. S. a. 1.

The Honey-Cherry.

Cerasus, sylvestris, septentrionalis, fructu rubro parvo serotino. Raii.
Syn.

L. 1. CH. 3.

II.

CL. 1. S. a. 2.

THE JULY ARRACH MOTH. The *Caterpillar* may be found, full fed, on the *wild Arrach*, about the Middle of *September*; at which Time it

changes to a *Chrysalis* in the Earth: The *Moth* is bred in *July*, and may be taken in an Evening in and near the Gardens about *Vauxhall* and other Places.

See *Albin*, Pl. 47.



L. I. ch. III.

Cl. 1. S. a. 2.

The Black Curran-Tree.

Ribes nigrum vulgo dictum, folio olente. I. B.

L. 1. CH. 3.

III.

CL. 1. S. a. 3.

THE MOTTLED UMBER-MOTH. The *Caterpillar* feeds on Fruit-Trees, *Oak*, *Elm*, *Black-Thorn*, &c. and turns to a *Chrysalis*, within the Earth, about the Middle of *May*. The *Moth* comes forth in *October*, and may be found at that Time sticking on the Bark of Trees in Parks, &c.

See *Albin*, P. 100.



L. 1. ch. III.

Cl. 1. S. a. 3.

The Apricock-Plumb.

Prunus fructu parvo præcoci. Tourn.

L. 1. CH. 3.

IV.

CL. 1. S. a. 4.

THE JULY SALLOW MOTH. The *Caterpillar* feeds on the common *Sallow*, in the Top-leaves of which (spun together) it lies concealed in the Day-time. Some Leaves, thus spun together, looked as if they had been dead; but, on Examination, *Caterpillars* of this Sort were found therein: They changed to a *Chrysalis*, spun amongst the Leaves, at the End of *May*. The *Moth* is bred at the Beginning of *July*, and may be taken, in the Evening, by the Sides of Woods, Hedges, &c.



L. 1. ch. III.

Cl. 1. S. a. 4.

The Wild Black Cherry.

Cerasus major ac sylvestris, fructu subdulci nigro colore inficiente. C. B.
Pin.

L. 1. Ch. 3.

V.

CL. 1. S. a. 5.

THE BUFF ARGUS-MOTH. The two *Caterpillars* exhibited in this Plate were taken by me upon the *Oak*, they feed likewise on the *wild Briar*, *White-Thorn*, &c. They tied themselves up against the Leaves, after

the Manner of the *White Butterfly*, with their Heads upwards; changed into *Chrysalis* about the Middle of *June*, and the *Moths* were bred about the Middle of *July*. These I suppose to be Male and Female of the same Species. You may take them flying, in an Evening, by the Sides of Woods and Hedges, and in green Lanes.



L. I. ch. III.

Cl. 1. S. a. 5.

The double-flowered Sweet Briar.

Rosa sylvestris odora, sive eglanteria, flore duplice. Park. Parad.

L. 1. CH. 3.

VI.

CL. 1. S. a. 6.

THE GREEN BROOM-MOTH. The *Caterpillar* feeds on the Leaves of the common *Broom*; and, about the Middle of *June*, changes to a *Chrysalis* in a thin Spinning; in which State it continues for about three Weeks, then the *Moth* appears, and may be taken in the Day-time, by beating the *Furze* and *Broom*, wherein they usually are found.

See *Rosel*, Cl. 3. Tab. 12.



L. I. ch. III.

Cl. 1. S. a. 6.

The Needle Furze.

Genista minor Aspalathoides, vel Genista spinosa Anglica. C. B. Pin.

L. 1. CH. 3.

VII.

CL. 1. S. a. 7.

THE LIME MOSS MOTH. The Eggs are of a greenish Colour, and are laid in *June*; after being in that State about ten Days the *Caterpillars* are hatch'd. They feed on the *Moss* that grows on *Lime* Trees; live and feed all the Winter in the *Caterpillar* State, and about the Beginning of *May*, are full grown, at which Time they make a Spinning amongst the *Moss*, and change to *Chrysalis*. The *Moths* are bred at the Beginning of *June*, and may be taken in the Evenings, by the Sides of Woods, in green Lanes, &c.



L. I. ch. III.

Cl. 1. S. a. 7.

The Yellow-flowered Sage.

Phlomis fruticosa, *Salviæ folio latiore & rotundiore*. Raii Hist. 511.

SECT. *b.*

Naked Loopers, *with* Protuberances.

L. 1. CH. 3.

I.

CL. 1. S. b. 1.

THE SPOTTED ELM-MOTH. Its *Caterpillar* is not very common, altho' it feeds on the Leaves of the *Elm*, *Lime* and *Oak*. It goes into the Earth, and turns to a *Chrysalis* in *August*: The *Moth* appears in *May*, and is then to be found sometimes, sticking on the Barks of the above Trees; but it is not common.

See *Albin*, P. 40, 41. 91, 92.



L. I. ch. III.

Cl. 2. S. b. 1.

The Witch Elm.

Ulmus folio latissimo scabro. Ger. Em. 1481.

L. 1. CH. 3.

II.

CL. 1. S. b. 2.

THE SWALLOW-TAIL MOTH. The *Caterpillar* feeds on *Bramble*, *Oak* Leaves, and other Growths; lives all the Winter: When it changes to a *Chrysalis*, which it does in *March*, the *Chrysalis* is fastened by the Tail, and lies, with its Head upwards, within a few fine Threads of its own spinning. The *Moth* is bred in *May* and *June*, and may be taken flying in an Evening by the Sides of Woods, and in green Lanes.

See *Rosel*, Cl. 3. Tab. 6. *Albin*, P. 94.



L. 1. ch. III.

Cl. 2. S. b. 2.

The Bramble.

Rubus, fructu nigro. C. B.

L. 1. CH. 3.

III.

CL. 1. S. b. 3.

THE OCTOBER-MOTH. Its *Caterpillar* is to be found, full fed, at the End of *May*, by beating the Boughs of the *Oak, Elm, Black-Thorn, &c.* which are the Food it feeds upon. It changes to the *Chrysalis* State, within

the Earth, about that Time, and the *Moth* is bred in *October*, when it may be found sticking on the Bark of Trees, in Parks and other Places.



L. I. ch. III.

Cl. 1. S. b. 3.

The smooth leaved Elm.

Ulmus folio Glabro. Ger. Em. 1481.

L. 1. CH. 3.

IV.

CL. 1. S. b. 4.

THE BRIMSTONE MOTH. The *Caterpillar* is found on the *White-Thorn*, towards the Middle of *September*, when it is full fed, and changes to a *Chrysalis*, within a *Spinning*, and the *Moth* is produced in the *April* and *May* following. It flies slowly, and is easily taken in an Evening. This *Moth* breeds twice in a Year; the first Time in *April* and *May*, the second in *July* and *August*.



L. I. ch. III.

Cl. 1. S. b. 4.

Figure 1. The White Thorn. 2. The Blue Bells. 3.
The Red Ranunculus.

Fig. 1. *Oxyacanthus*. Ger. 2. *Hyacinthus Anglicus*. Ger. 3. *Ranunculus hortensis flore rubello*.

L. 1. CH. 3.

V.

CL. 1. S. b. 5.

THE SCALLOP WINGED MOTH laid her Eggs on the 5th of *August*, and eight Days after the *Caterpillars* appeared. I fed them with *White-Thorn* and *Privet*, till the Beginning of *October*, at which Time several of them changed to the *Chrysalis* betwixt the Leaves spun together. They remained in that State during the Winter, and the *Moths* were bred in *May* following. This *Moth* is to be taken, in the Evening, by the Sides of Woods, Hedges, &c.



L. 1. ch. III.

Cl. 1. S. b. 5.

The White Thorn, with its Fruit.

Mespilus, Apii folio; sylvestris, sive oxyacantha. C. B. Pin. 458.

L. 1. CH. 3.

VI.

CL. 1. S. b. 6.

THE RICHMOND-BEAUTY, a MOTH. Mr. *Rosel* assures us, that the *Caterpillars* feed on the Leaves of the *Pear* Tree, &c. and is to be found at the End of *May* and the Beginning of *June*. They change to the

Chrysalis amongst the *Pear* Leaves: They continue fourteen Days in that State, and the *Moth* appears about the End of *June*. This *Moth* is to be taken in an Evening by the Sides of Woods, in green Lanes, Gardens, &c.

See *Rosel*, Cl. 2. Tab. 10.



L. I. ch. III.

Cl. 1. S. b. 6.

The *Spanish* or *Catalonian* Jasmine.

Jasminum humilium magno flore. C. B. Pin.

THE HAWTHORN MOTH. The *Caterpillars* that produce this Kind of *Moth* are of various Colours, some being much lighter than others. I got several of them by beating the Boughs of the *White Thorn*, about the Middle of *September*; at which Time many of this Sort spin themselves amongst the Leaves and change to *Chrysalis*, and the *Moths* are bred in the Middle of *May* following. This is a very scarce Fly.



L. 1. ch. III.

Cl. 3. S. b. 7.

The Carolina Hawthorn.

Mespilus Caroliniana, Apii foliis volgari similis major fructu luteo.
Millar.

L. 1. CH. 3.

VIII.

CL. 1. S. b. 8.

THE ORANGE MOTH. The four different coloured Flies represented in the Plate, were all produced from *Caterpillars* that were hatched from the Eggs of such a Female as is represented at *Fig. 1*. The Eggs were of a pale green Colour, and were hatch'd on the 13th *July* 1747; and the young *Caterpillars* were fed with the Leaves of *Black-Thorn*, till about the 20th, when they changed their first Skin; on the second of *August* they changed their Skin again, and, about the Middle of the same Month, they left off eating: But, *October* the 25th, being put on *Bramble* Leaves, they began to eat again, and changed their Skin a third Time, *November* the 28th; after which they remained pretty quiet the whole Winter without eating. At the End of *March* they began to eat the fresh Buds of the *Black* and *White-Thorn*, and changed their Skin for the fourth Time. About the End of *April*, one that was separated from the Rest changed its Skin a fifth Time *May* the 10th: spun up *May* the 29th: and the *Moth* was produced *June* the 17th. Several others spun up between *May* the 22d and *June* the 5th, and the *Moths* were produced between *June* the 12th and 25th. One that did not change its Skin, for the last Time, till *June* the 26th, was given to the Proprietor of this Work; it spun up in the Leaves, on the 10th of *July*, and the *Moth*, *Fig. 2*. was bred on the 1st of *August*. Of eight *Moths* that were produced, three were Females of the common Orange Kind, and five were Males, one of which only was all over of an Orange Colour. Amongst the *Caterpillars* there was one that changed its fourth Skin, *July* the 4th, and fed for some Time; however, it has not yet changed its fifth Skin, tho' it continued alive, without eating, till *December* the 25th 1748, at which Time it was put on *Bramble* Leaves, and has continued to feed thereon till the present 3d of *February* 1748-9.—The Author of this Work was favoured with the preceeding Account by a most skilful and exact observer, whose Name, had he leave to mention it, would do him great Honour.



L. I. ch. III.

Cl. 1. S. b. 8.

The Dwarf-Willow.

Salix pumila Alpina, rotundifolia repens, inferne subcinerea. C. B. Pin.



C L A S S II.

Hairy Loopers.

S E C T . *a*.

Having no Protuberances.

L. 1. CH. 3.

I.

CL. 2. S. *a*. 1.

THE LARGE MAGPYE, or CURRAN MOTH. Most Gardens and Hedges abound with the *Caterpillars* of this Fly, which feed on *Curran* and *Gooseberry-Bushes*, and also on the *Black Thorn*, &c. They change into the *Chrysalis State*, hanging by the Tail, at the End of *May*, and the *Moths* appear in the Middle of *June*. They are very common, and easily taken in an Evening in the Gardens, and most other Places, and may likewise be catch'd in the Day-time, by beating the Hedges and Bushes. The *Caterpillar* lives all Winter, and in the Spring begins to feed as soon as the Buds are open.

See *Lister* on *Godart*, P. 13. N. 9. *Merian*, Vol. 1. Chap. 29. *Albin*, P. 43. *Rosel*, Class 3. Tab. 2.



L. 1. ch. III.

Cl. 2. S. 1. a. 1.

The Curran Tree.

Ribes major fructu rubro, Hort. Eyst.

L. 1. CH. 3.

II.

CL. 2. S. a. 2.

THE GOOSEBERRY-MOTH. The *Caterpillars* are hatch'd at the latter End of the Year, and live all Winter. About the Middle of May you may find Plenty of them on the *Gooseberry* and *Curran* Bushes; about which Time they go into the Ground, and turn to *Chrysalis*, and about the

Middle of *June* the *Moths* appear. They are common in most Gardens in the Evening, and may be taken in the Day-time, as the last.

See *Lister* on *Godart*, P. 19. N. 12. *Merian*, Vol. 1. Chap. 25. *Albin* P. 47. *Rosel*, Class 3. Tab. 4.



L. I. ch. III.

Cl. 2. S. a. 2.

The Gooseberry Tree, with its Flower and Fruit.

Grossularia, fructu obscure purpurascente. Clus.

Hairy Loopers.

S E C T . *b.*

Having Tufts of Hair, partly round each Joint.

L. 1. CH. 3.

I.

CL. 2. S. *b.* 1.

THE BRINDLE-MOTH. This *Caterpillar*, which is not common among the Loopers, was found by Mr. *Dandridge* on the *Hasle*, at the Beginning of *June*. It also feeds on *Oak*, &c. On the 14th of the same Month it went into the Ground and changed into a *Chrysalis*, and at the Beginning of *April* following the *Moth* came out. The Females have no Wings. I have had several of this Kind of *Moth* bred in *February* from the *Chrysalis*, which I got by digging with a Trowel at the Root of Oak Trees. The *Moth* is rarely taken in the Fly-state.



L. I. ch. III.

Cl. 2. S. b. 1.

Bladder-Sena, *with* Reddish-Seed Vessels.

Colutea vesicaria, vesiculis rubentibus. J. B.



C H A P. IV.

Caterpillars shaped like Wood-Lice.

C L A S S I.

S E C T. *a.*

L. 1. CH. 4.

I.

CL. 1. S. *a.* 1.

THE SMALL OAK-EGGER-MOTH. This *Caterpillar* I took in *Hornsey-Wood* on an Oak-Tree, at the Beginning *October*, soon after which Time it chang'd to a *Chrysalis*, by fastening the Oak-Leaf close to the Bottom of the Box I kept it in with a Spinning. Thus it remained through the Winter, and the *Moth* was bred at the End of *May*. Both *Caterpillar* and *Fly* are very rare.



L. I. ch. IV.

Cl. 1. S. a. 1.

The Double Velvet-Rose.

Rosa holosericea, multiplex. Park. Parad.



C H A P. V.

Of Moths, whose Generation is not yet known.

C L A S S I.

S E C T. *a*.

L. 1. CH. 5.

I.

CL. 1. S. A. 1.

THE GLORY OF KENT, a *Moth*. This is the only *Moth* of the Kind that I have yet heard was ever taken in *England* or elsewhere. It was found about the Middle of *April*, 1741, flying in a Wood, in the Day-time, near *Cookham*, by *Westram*, in *Kent*. It was taken by Mr. *William Constable*, who lives near that Place; and the *Moth* is at present in the Collection of Mr. *Charles Cabrier*, of *London*.



L. L. ch. v.

Cl. 2. S. A. 1.

The Double-Flowering Peach.

Persica Africana nova, flore incarnato pleno. Tourn.

L. 1. Ch. 5.

II.

CL. 1. S. A. 2.

THE CLEIFDEN NONPAREIL, a *Moth*. This curious Fly was found by Mr. *Davenport*, sticking against the Body of an *Ash Tree*, near *Cleifden*, in *Buckinghamshire*. It was taken in the Month of *July*: Is at

present in the Possession of *Charles Lockyer*, Esq; and is the only one of the Sort that I have yet seen or heard of.



L. L. ch. v.

Cl. 1. S. A. 2.

The Ciphian Rose.

Rosa; pimpinella minor Scotica, floribus ex albo & corneo eleganter variegatis. Pluk. Alm.



B O O K II.

C H A P. I.

Caterpillars producing *Flies*, which resemble partly the *Moth*, and partly the *Butterfly*.

C L A S S I.

S E C T. *a*.

L. 2. CH. 1.

I.

CL. 1. S. *a*. 1.

THE BURNET-MOTH. The *Caterpillars* are to be found feeding on *Hay-Grass* about the Middle of *May*, at which Time many of them go into the *Chrysalis* State, within such a Spinning as is expressed in the Plate, and which they make fast to the Stalks of Grass, about a Foot from the Ground. The *Moths* fly about the Middle of *June*, in the Day-time, and are easily taken.

See *Albin*, P. 82. *Rosel*, Cl. 2. Tab. 57. Vol. 1. *Reaumur*, Pl. 12. Pag. 284. Vol. 2. Pl. 2. Pag. 120.



L. II. ch. I.

Cl. 1. S. a. 1.

The Burnet Rose.

Rosa pumila, spinosissima, foliis pimpinellæ, glabris, Flore albo. J. B.



B O O K III.

C H A P. I.

CATERPILLARS, *whose Bodies are* SMOOTH, *producing* BUTTERFLIES.

C L A S S I.

S E C T. *a.*

CATERPILLARS *without any* Protuberances, *whose* Chrysalis *is fixed by the*
Tail, and hangs in an Horizontal Position, by Means of a Thread fastened
round the Back and on each Side, and producing SWALLOW-TAIL
BUTTERFLIES.

L. 3. CH. 1.

I.

CL. 1. S. *a.* 1.

THE SWALLOW-TAIL BUTTERFLY is produced twice a Year. The first Brood appears in *May*, the second towards the End of *July*. Being in a Meadow near *Cookham*, in *Kent*, on the 5th Day of *August*, 1748, I observed a Female *Swallow-Tail* hovering over certain Plants, which taking particular Notice of, I found to be the *Meadow Saxifrage*, and examining them carefully, I discovered four Eggs just laid by the *Fly*, wherewith I was highly pleased. On the 13th of the same Month these Eggs produced *Caterpillars*, of Size and Colour as expressed at Figure 1. On the 19th one of the *Caterpillars* shifted its Skin, and appeared like Figure 2. On the 23d it changed again to the Likeness of Figure 3. On the 4th of

September it became like Figure 4. On the 11th of the same Month it appeared in its fifth and last Skin, which was extreamly beautiful, as expressed at Figure 5. On the 22d the *Caterpillar* was full grown, and fixed itself in the Manner of Figure 6, in order to change into the *Chrysalis* 7, which was produced on the 26th of *September*; in which State it remains, *January* 20, 1748-9. I fed the *Caterpillar* from its being first hatched with the green Leaves of the common Carrot, which it eat plentifully. This fine *Butterfly* may be taken in the Meadows and Clover Fields about *Cookham*, near *Westram*, in *Kent*, at the Times above-mentioned. It flies so swiftly that it is in vain to follow it, you must therefore wait 'till it settles, and then, if you are near, be nimble, and you may catch it without much Difficulty. The *Flies* are represented in the next Plate numbered the same as this.

See *Reaumur*, Pl. 29. Pag. 486. Also Pl. 30. Vol. 1. Pag. 486.



L. III. ch. I.

Cl. 1. S. a. 1.

The Meadow Saxifrage.

Seseli pratense nostras. Park.



L. III. ch. I.

Cl. 1. S. a. 1.

L. 3. CH. 3.

II.

CL. 1. S. a. 2.

THE BRIMSTONE-BUTTERFLY lives in the *Fly* State all the Winter Season, for very early in the Spring you will see them in the Woods flying about and seeking out their Mates. The Food of the *Caterpillar* is reported by Mr. *Dandridge* to be the *Buck-Thorn*, on which he fed it 'till the Middle of *June*, when it changed to a *Chrysalis*, and the *Fly* was bred at the Beginning of *August*. The *Caterpillars* are rare to be met with, although the

Fly is common in Woods and Fields of Clover, in the Months of *August* and *September*.



L. III. ch. I.

Cl. I. S. a. 2.

The Buck-Thorn, with its Berries.

Rhamnus Catharticus. C. B. P.

C L A S S II.

SECT. *a.*

CATERPILLARS *having little or no* HAIR.

Fastening their Chrysalis as the last Sort, and producing ROUND-WINGED
BUTTERFLIES.

L. 3. CH. 1.

I.

CL. 2. S. *a.* 1.

THE WHITE-BUTTERFLY, with BLACK VEINS. The *Caterpillars* that produce this sort of *Butterfly* are sociable and feed together on the *White-Thorn*, till about the Middle of *May*, at which Time they are full fed and change to the *Chrysalis*. The *Fly* is bred in *June*, and is common among *Hay-Grass*.

See *Merian*, Vol. 2. Chap. 35. *Moss*. P. 103. *Raii*, Hist. Insect. P. 115, N. 5. *Albin*, Plate 2. *Rosel*, Cl. 2. Tab. 3. *Reaumur*, Vol. 2. Pl. 2. P. 120.



L. III. ch. I.

Cl. 2. S. a. 1.

The Mountain-Elder, and White-Thorn.

Sambucus, racemosa rubra. C. B. P.

L. 3. CH. 1.

II.

CL. 2. S. a. 2.

THE LARGE WHITE GARDEN-BUTTERFLY breeds twice a Year. The *Caterpillars* thereof are well known in Gardens and other Places. Their first Brood is in *May*, and the second about two Months after.

The *Chrysalis* that produces the second Brood is only fourteen Days in that State, and the *Fly* is bred in *July*. Those *Caterpillars* that go to *Chrysalis* about *September* remain in that State all the Winter, and produce their *Flies* in *May*. These *Caterpillars* are great Devourers of *Cabbage* and *Colliflowers* from *June* to *September*, which last Month is the Time of their changing into the *Chrysalis* State. In order to preserve your Cabbages, &c. 'tis worth while to destroy their *Chrysalides* in the Winter, when they may commonly be found under the Copping of Garden Walls, Pales, or any other Places a little shelter'd: Their Eggs likewise should be sought after, and may be found in great Numbers together on the under Sides of the Leaves. *Note*, The *Swallow-Tail Butterfly* is always bred at the same Time as this *Fly* is.

See *Godart*, Sec. 1. Num. 7. *Merian*, Vol. 1. Chap. 45. *Moufet*, P. 103. *Raii*, Hist. Insect. p. 113. *List. on God.* P. 16. Fig. 7. *Albin*, Plate 1. *Rosel*, C. 2. Tab. 4.



L. III. ch. I.

Cl. 2. S. a. 2.

The Nasturtium Flower.

Acriviola. J. K. H.

L. 3. CH. 1.

III.

CL. 2. S. a. 3.

THE SMALL WHITE GARDEN-BUTTERFLY breeds twice a Year: The first Brood is at the Beginning of May, and the second in July. The Caterpillar feeds on Cabbage, Horse-Radish, Cresses, &c.

Towards the End of *June* they tie themselves up by the Tail with a Thread, which goes likewise cross their Middle. The *Fly* comes forth in *July*, and is very common.

See *Albin*, P. 51. *Rosel*, Cl. 2. Tab. 5. *Lister* on *Godart*, P. 12. N. 8. *Reaumur*, Vol. 1. Pl. 29. Pag. 486. Vol. 2. Pl. 2. Pag. 120.



L. III. ch. I.

Cl. 2. S. a. 3.

The Double-Flowering Almond.

Amygdalus sativa flore pleno. J. R. H.

L. 3. CH. 1.

IV.

CL. 2. S. a. 4.

THE WHITE-BUTTERFLY with GREEN VEINS. This also breeds twice a Year: Its first Brood is in *May*, the second in *July*. The *Caterpillar* feeds on *Cabbage*, &c. like the foregoing, ties itself up in the same Manner, and the *Fly* is bred in *July*. It is very common, and may be taken in most Gardens.

See *Albin*, P. 52. *Merian*, Vol. 2. Chap. 39.



L. III. ch. I.

Cl. 2. s. a. 4.

The Purple Auricula.

Auricula, ursi flore Purpureo. Lob.

L. 3. CH. 1.

V.

Cl. 2. S. a. 5.

THE ORANGE-TIP, or LADY OF THE WOODS, A BUTTERFLY. We are obliged to Mr. *Rosel* for the Discovery of the Progress of this and many other curious *Flies*. He says the *Caterpillar* feeds on the *Brassica Sylvestris*, or *Wild Cole*, whereon the *Fly* lays its Eggs. The *Caterpillar* is found in *June* and *July*, at which Time it changes into a *Chrysalis*. It remains in that State through the Winter, and the *Fly* is produced in *May*. It is very common in Fields and green Lanes, and breeds but once a Year.

See *Rosel*, Cl. 2. Tab. 8.



L. III. ch. I.

Cl. 2. S. a. 5.

The Mountain-Curran, and the Wild Cole.

Ribes Alpinus dulcis. J. B. Brassica Sylvestris.



CATERPILLARS, having little or no Hair, producing SCALLOP-WINGED
BUTTERFLIES.

SECT. *b*.

L. 3. CH. 1.

I.

CL. 2. S. *b*. 1.

THE MARMORIS, or MARBLE BUTTERFLY. On the 8th of *July*, 1748. I took with my Net several Females of this sort of *Fly*, which is very common amongst the *Hay-Grass*. I put them in a Box, taking Care to hurt them as little as possible, with full Hopes that some of them would lay their Eggs; and according to my Wishes the next Morning I had 160 Eggs, which, when first laid, were of a fine yellow Colour, but in less than a Minute's Time they became perfectly white, and are not glutinous like many other Sorts. On the 27th and 28th of the same Month the young *Caterpillars* appeared, and were put on common Grass to feed: They continued extremely small for some Months, and, notwithstanding great Care was taken of them, only three *Flies* were produced, the rest all dying in the *Caterpillar* State.

The different Changes observed in these three were as follows.

Caterpillars changed their Skins.	Went to <i>Chrysalis</i> .	The <i>Flies</i> were bred.
{ 1st April, 11 May 6 May 26	June — 14	July — 11 }
The { 2d April, 28 May 21 June 6	June — 30	July — 23 } 1748
{ 3d May, 1 May 21 June 9	July — 4	July — 27 }

The *Caterpillars* that produced these *Flies* changed to *Chrysalis* on the Ground, without fastening themselves to any Thing, and were fed all the Time with common Grass, and it is presumed changed their Skins twice before the 11th of *April*.



L. III. ch. I.

Cl. 2. S. b. 1.

Figure 1. The Bent Grass. 2. Brown Millet-Grass. 3. Yellow Ladies Bedstraw. 4. The Great Fox-tail Grass.

1. *Gramen segetum, panicula speciosa*. Park. 2. *Gramen miliaceum majus panicula spadicea*. Petiv. 3. *Gallium luteum*, C. B. 4. *Gramen Alopecuroides majus*. Ger.

L. 3. CH. 1.

II.

CL. 2. S. b. 2.

THE MEADOW BROWN-BUTTERFLY. The *Caterpillar* was found by Mr. *Dandridge* on the common Grass, with which he fed it 'till the 24th of *May*, when it fastened itself up by the Tail, and changed into a short thick green *Chrysalis* streaked with black, and on the 11th of *June* came out the *Butterfly*, which is one of the most common sorts the Fields afford, and yet the *Caterpillar* is rarely found.

See *Albin*, P. 53.



L. III. ch. 1.

Cl. 2. s. b. 2.

The Water Elder.

Opulus Ruellii. Instit. R. Herb.

L. 3. CH. 1.

III.

CL. 2. S. b. 3.

THE GREAT ARGUS-BUTTERFLY breeds twice a Year: The first Time is in *May*, the second in *July*. *Merian* says, the *Caterpillar* feeds on *Grass*, and that it changes to a small green *Chrysalis*, from which proceeds this elegant *Fly*, adorned with brown and black Spots, having two white Horns spotted with black, and Eyes of a beautiful Colour. This *Butterfly* is very common; its Flight is not swift: It delights to settle on dry Banks and Walls, and in Path-Ways, and is not difficult to take. Its *Caterpillar* is seldom found.

See *Merian*, Vol. 2. Chap. 4.



L. III. ch. I.

Cl. 2. S. b. 3.

The Passion-Flower.

Granadilla Pentaphyllos Flore cæruleo magno. Boerh.

L. 3. CH. 1.

IV.

CL. 2. S. b. 4.

THE WOOD ARGUS-BUTTERFLY. This also breeds twice a Year: The first Brood appears at the Beginning of *May*, the second at the Beginning of *July*. It flies slowly, and is to be taken in Woods, green Lanes,

&c. *Reaumur* informs us, that the *Caterpillar* feeds on *Grass*; that it changed to a *Chrysalis* on the 19th of *June*, and that ten Days after the *Fly* was bred.

See *Reaumur*, Plate 27. Pag. 431.



L. III. ch. I.

Cl. 2. S. b. 4.

The Spanish Broom.

Genista Iunca, J. B.

SECT. C.

CATERPILLARS *having little* HAIR, *producing* BUTTERFLIES, *whose Head and Body are much larger (for their Size) than any other* BUTTERFLIES: *Of which there are several belonging to this Section, whose Caterpillars have not as yet been discovered.*

L. 3. CH. 1.

I.

CL. 2. S. c. 1.

THE GRIZZLED BUTTERFLY. Mr. *Rosel* tells us, that the *Caterpillar* of this *Fly* was found on the *Mallow*, with the Leaves of which he fed it till the End of *June*, when it spun a Web amongst the Leaves, and changed to a *Chrysalis*, the *Butterfly* of which was bred the *May* following. This *Fly* is to be taken in Woods and Meads, at the Beginning of *May*: and although small, it flies swiftly, so that you must be very quick to take it.

See *Rosel*, Cl. 2. Tab. 10. *Merian*, Vol. 1 Chap. 48.



L. III. ch. I.

Cl. 2. S. c. 1.

The common Mallow.

Malva vulgaris. Park.



C L A S S III.

C A T E R P I L L A R S armed with Spikes.

S E C T . a.

Whose Chrysalis hangs by the Tail perpendicularly downwards.

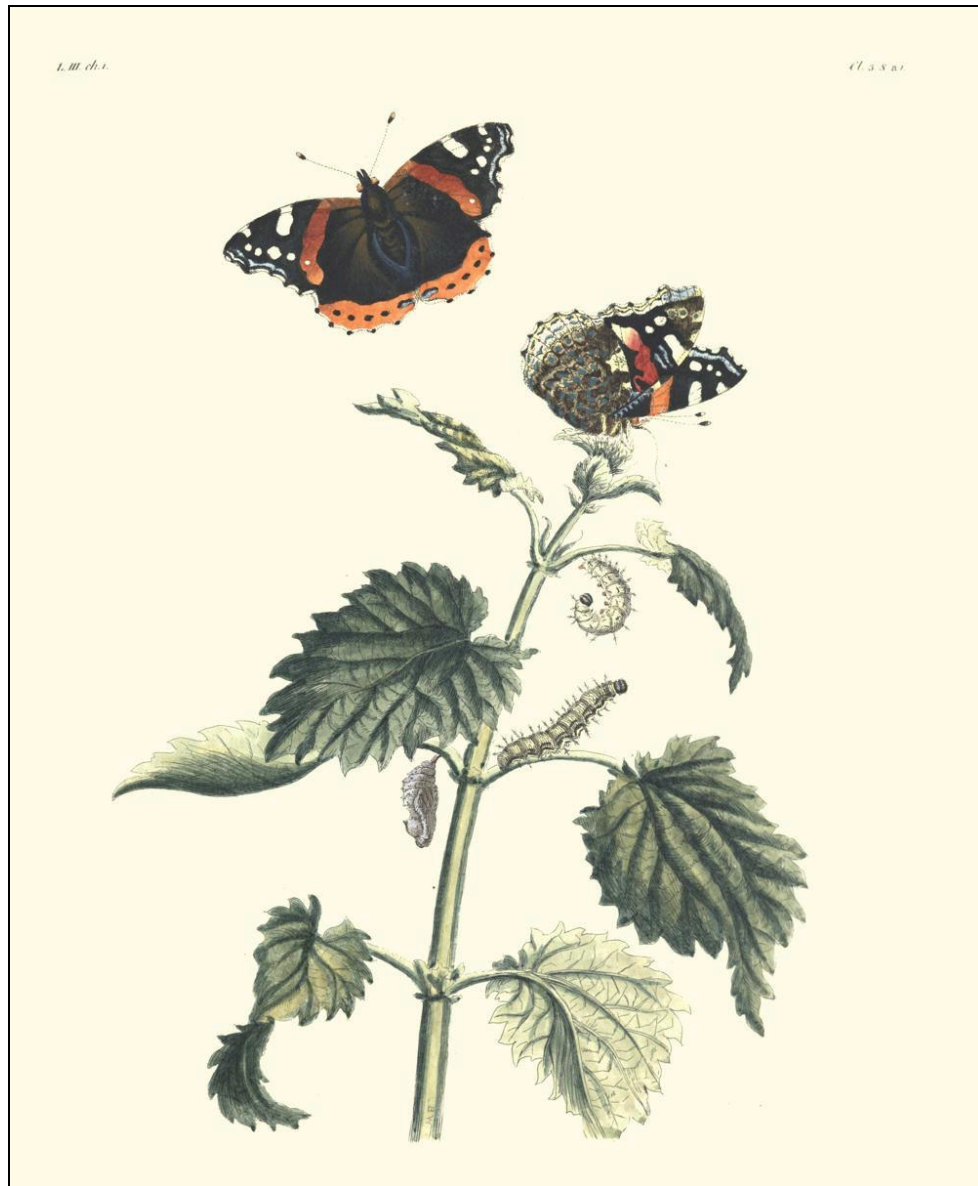
L. 3. CH. 1.

I.

CL. 3. S. a. 1.

THE ADMIRABLE BUTTERFLY . The great *Stinging Nettle* is the Food of this *Caterpillar*, which is to be found by observing where the *Nettle-Leaves* appear to be folded or spun together, for within such Leaves the *Caterpillar* feeds. It turns in *August* to a *Chrysalis*, hanging perpendicularly downwards by the Tail, fourteen Days after the *Chrysalis* is formed the *Fly* appears. The *Caterpillars* shelter themselves after this Manner, that they may be the better secured from the too great Heat of the Sun, from Rain, Birds, and a small *Ichneumon Fly*, which often hinders their coming to Perfection, by laying its Eggs therein, which Eggs are of such a glutenous Nature, that they stick fast as soon as laid. Some of the *Chrysalides* appear as if gilded with burnished Gold, but such usually produce not a *Butterfly*, but a Brood of small, though very beautiful *Ichneumons*. The *Fly* may be taken in Gardens and other Places, it feeds on *Fruit* that lies under Trees, &c. and is no uncommon *Fly*. It lives quite through the Winter.

See *Goedart*, Vol. 1. Tab. 26. Pag. 96. Fig. Opt. List. Fig. 4. Gr. V. 2. Pag. 81. Tab. 81. Fig. Opt. Moff. 100. Num. 6. *An. Hoef.* Tab. 12. Fig. 15. *Raii*, Hist. Insect. P. 126. *Albin*, Pl. 3. *Reaumur*, Pl. 10. Pag. 284, Vol. 1.



L. III. ch. 1.

CL. 3. S. a. 1.

The Great Stinging-Nettle.

Urtica major vulgaris. J. B.

L. 3. CH. 1.

II.

CL. 3. S. a. 2.

THE PEACOCK-BUTTERFLY. You must look for the *Caterpillar* that produces this *Fly* in the great *Stinging-Nettle*. They are sociable and feed together, are very easy to find, and may be taken full fed

towards the End of *June*, at which Time many of them change to the *Chrysalis*, and about fourteen Days after the *Fly* appears. It is very common, and may be taken on *Thistles*, *Burdock*, *Clover*, &c. The *Chrysalis* of this *Fly*, as well as the preceding, sometimes looks as if gilt with Gold: but such fine outsides usually produce *Ichneumons*. This *Fly* likewise lives all the Winter.

See *Reaumur*, Pl. 25. Pag. 446. Vol. 1. *Goed.* Vol. 1. P. 23. Fig. opt. 1. *List.* Fig. 1. *Graf.* Vol. 1. Tab. 26. Pag. 53. *Raii*, Hist. Insect. Pag. 122. N. 14. *Moufet*, P. 99. N. 4. *Hoef.* Tab. 12. Fig. opt. 9. *Johns.* Inst. 40. n. 4. Tab. 5. *Albin*, Pl. 4. *Rosel*, Cl. 1. T. 3.



L. III. ch. 1.

Cl. 3. S. a. 2.

Smith's Newington Peach.

L. 3. CH. 1.

III.

CL. 3. S. a. 3

THE PAINTED-LADY-BUTTERFLY. In this Plate two *Butterflies* are represented, the upper one of which is the *Painted Lady*. Its *Caterpillar* feeds on the great *Stinging-Nettle*, *Thistles*, &c. changes to a *Chrysalis* within the Leaves folded together in the Middle of *June*, is fourteen Days in that State, and then the *Fly* comes forth. It may be taken on the same Growths as the last, but is not so common.

Figure 4 represents, the SMALL TORTOISE-SHELL-BUTTERFLY, which is very common, and breeds twice a Year: The first Brood is towards the End of *June*, the second about the End of *August*. The *Caterpillar* feeds on *Nettles*, is open and sociable, and may be taken full fed about the Middle of *June*, and the Middle of *August*, and is about fourteen Days in *Chrysalis*. This *Butterfly* out-lives the Winter.

See *Lister* on *Goedart*, Pag. 7. N. 5. *Albin*, Pl. 56. Fig. 4. *Goed.* Vol. 1. Pag. 90, Fig. opt, 21. *List.* Fig. 2. *Graf.* Vol. 1. P. 89. *Mouff*, 101. N. 11, figured against N. 12. *Hoef.* Tab. 2. Fig. opt. 16. *Raii*, Hist. Insect. Pag. 117. N. 1. *Albin*, Pl. 4. *Rosel*, Cl. 1. Tab. 4. *Reaumur*, Pl. 26. Pag. 446. Vol. 1.



L. III. ch. 1.

Cl. 3. S. a. 3.

The Great Stinging-Nettle.

Urtica major vulgaris. J. B.

L. 3. CH. 1.

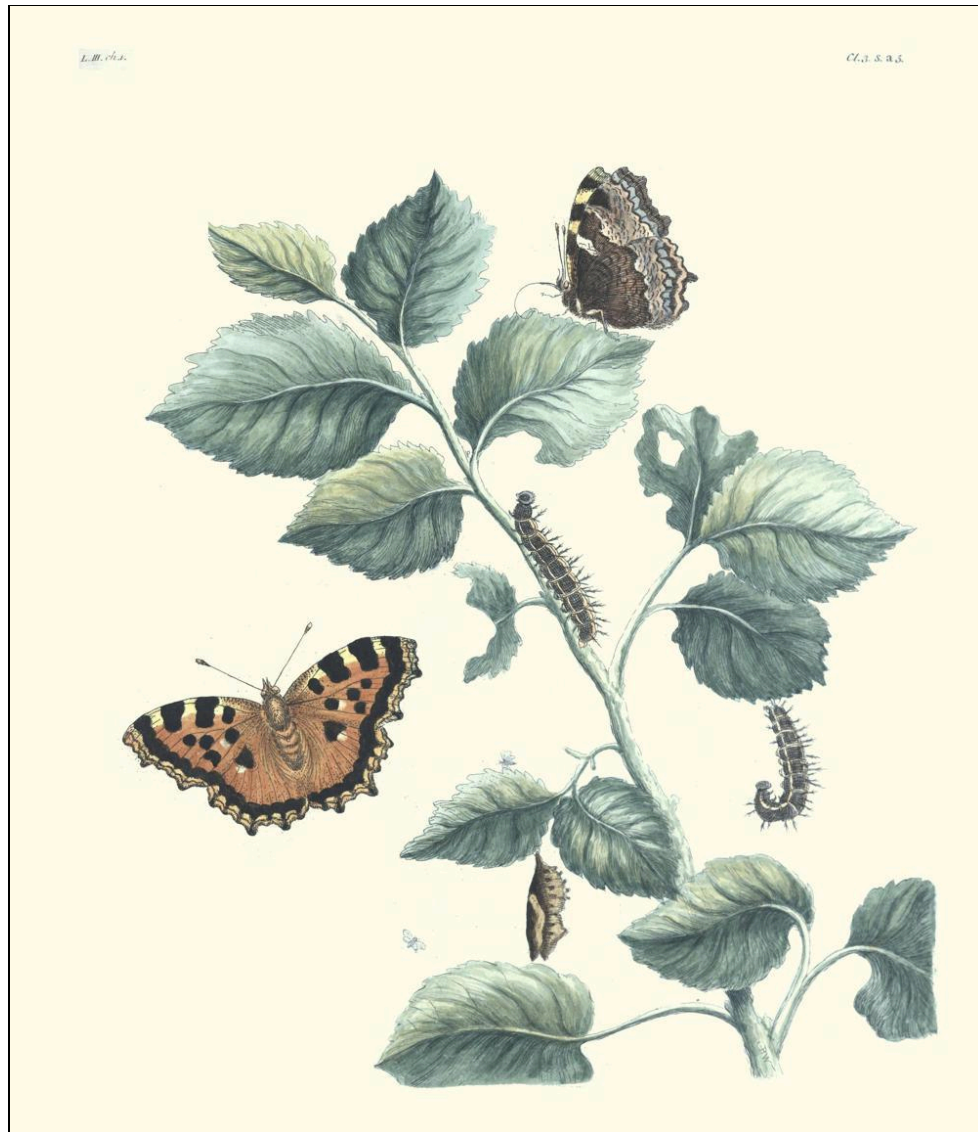
V.

CL. 3. S. a. 5.

THE GREAT TORTOISE-SHELL-BUTTERFLY. When the *Caterpillars* of this *Fly* are young, they feed together on the Leaves of the *Elm-Tree*, &c. About the Middle of *June* they are full fed, and usually

tie themselves up by the Tail under the Copings of Walls, or some such Shelter; in the Beginning of *July* the *Fly* comes out, and delights to settle in dry Path-ways, as also on the Bodies of Trees, &c. is swift in its Flight, and requires Nimbleness to take it. The *Chrysalis* is very apt to produce *Ichneumons* instead of its own *Butterfly*, those Creatures having laid their Eggs therein. This *Fly* lives all Winter.

See *Reaumur*, Pl. 23. Pag. 382. Vol. 1. *List.* on *Goedart*, Pag. 3. N. 3, *Merian*, Vol. 2. Tab. 2. *Albin*, Pl. 55. *Rosel*, Cl. 1. Tab. 2.



L. III. ch. I.

Cl. 3. s. a. 5.

The Common Elm-Tree.

Ulmus minor folio Angusto Scabro. Ger. Emac, 1480.

L. 3. CH. 1.

VI.

CL. 2. S. a. 6.

THE COMMA - BUTTERFLY breeds twice a Year: The first Time towards the End of *June*, and the second about the Beginning of *September*. The *Caterpillar* feeds on *Hop-Leaves*, *Nettles*, &c. puts on the *Chrysalis* Form in *June* and *August*, and is about fourteen Days in that State, when the *Fly* appears, which may be taken in Gardens, on *Blackberry* Blossoms, by the Sides of Hedges, and in such like Places. The Colours of the first Brood of this *Butterfly* are considerably lighter than as here represented.



L. III. ch. 1.

Cl. 3. S. a. 6.

The Double Wild Poppy, with a fiery-coloured
Flower edged with white.

Papaver, erraticum, flore pleno igneo marginibus candidis. H. L.

L. 3. CH. 1.

VII.

CL. 2. S. a. 7.

THE GREAT FRITILLARY-BUTTERFLY. Mr. *Rosel* says, the *Caterpillar* feeds on *Nettles*, in the private Recesses of Woods, that it

changes into the *Chrysalis* State at the End of *May*, and that the *Fly* is produced in *June*. This *Butterfly* is very swift in Flight, and is best taken in the Forenoon, when it will settle and feed on *Blackberry Blossoms*, *Thistles*, &c. They are most commonly found in Woods, and the Fields adjacent to Woods.

See *Rosel*, Cl. 1. Tab. 7.



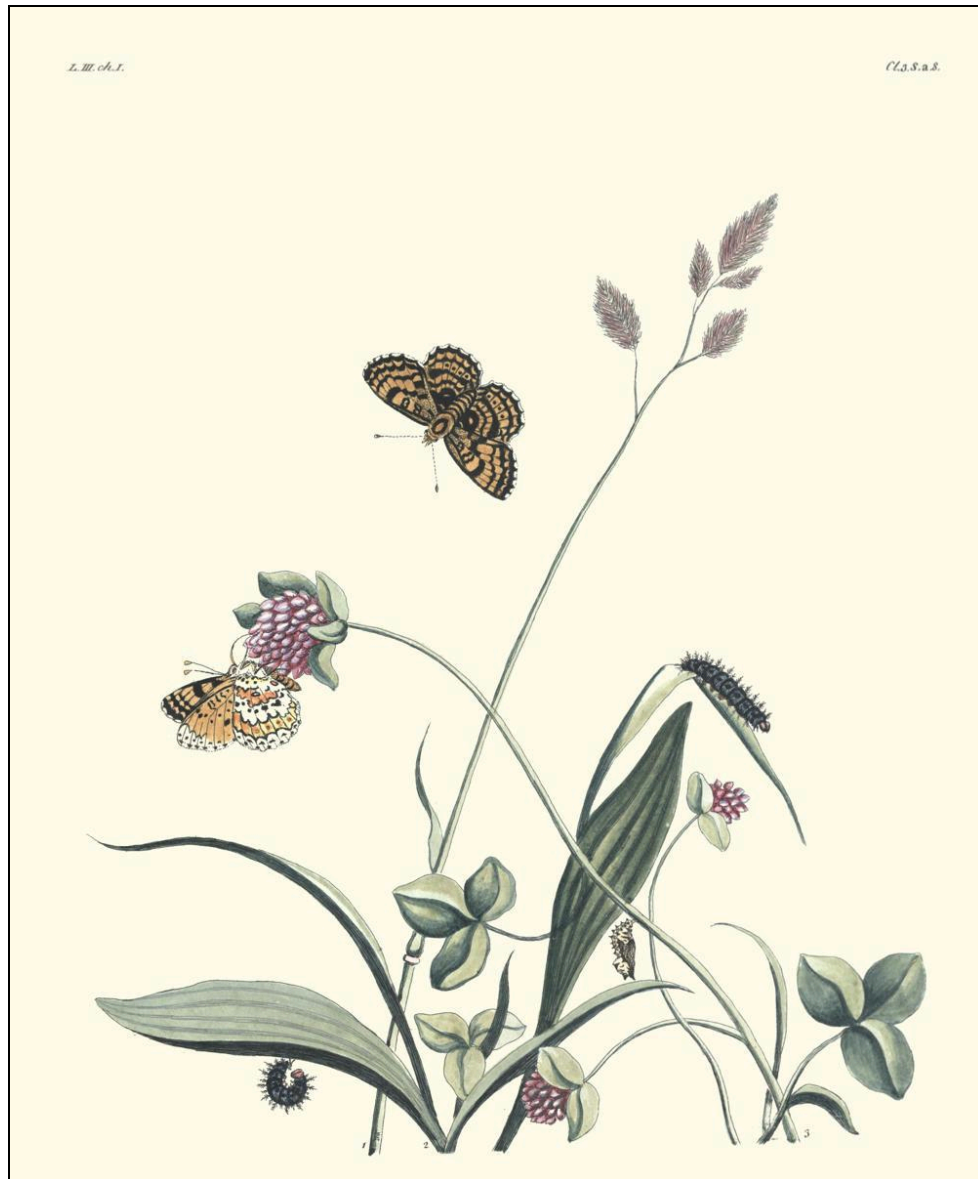
L. III. ch. 1.

Cl. 3. S. a. 7.

The Bramble.

Rubus major, fructu nigro. J. B.

THE PLANTAIN FRITILLARY-BUTTERFLY. The *Caterpillar* hereof feeds on *Plantain*, *Clover*, and *Grass*, changes to a *Chrysalis*, within a Web of its own spinning, upon the Surface of the Ground, at the Beginning of *May*, and the *Fly* appears fourteen Days after. The *Caterpillars* are sociable, and feed together. They appear of a very timorous Nature, for if you move the Food on which they are, they immediately quit their Hold and fall to the Ground, and there remain in a curl'd up Form till such Time they think the Danger over. The *Butterfly* is swift in Flight, but may be taken if diligently attended to, in Fields of *Hay-Grass*, at the Time above-mentioned.



L. III. ch. I.

Cl. 3. S. a. 8.

Figure 1. Rough Grass. 2. Plantain. 3. Clover.

1. *Gramen asperum*. J. B. 2. *Plantago major vulg.* C. B.

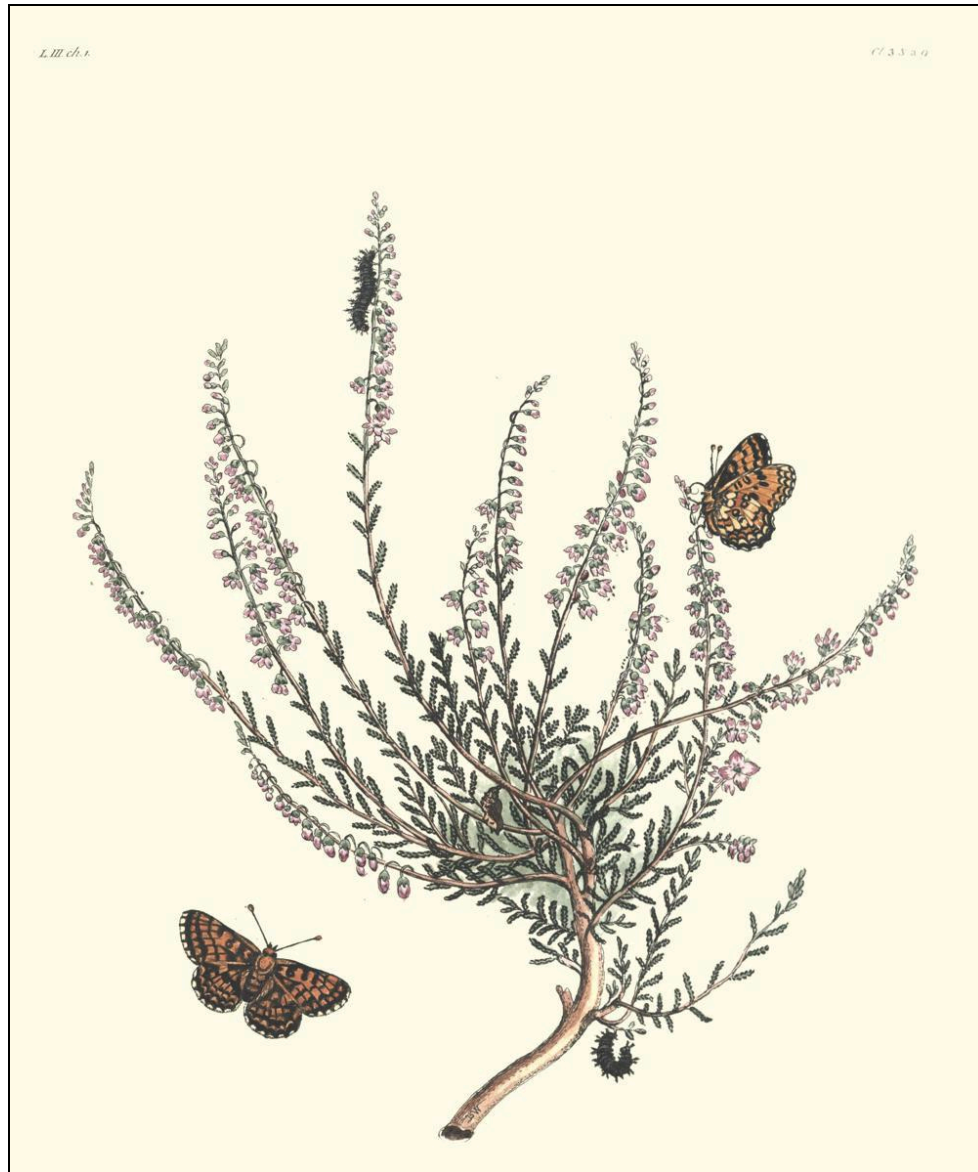
L. 3. CH. 1.

IX.

CL. 2. S. a. 9.

THE HEATH FRITILLARY-BUTTERFLY. I found the *Caterpillars* of this *Fly* feeding on common *Heath* in *Tottenham-Wood*, about the Middle of *May*, 1745, and they are of the same fearful Nature as

the last-mentioned. Six or seven of them were feeding near each other, I observed their Manner of eating, which was extremely quick, and when they moved it was at a great Rate. I fed them with common *Heath* for three or four Days; at the End of which some of them changed into *Chrysalis*, in which State they remained about fourteen Days, and then the *Flies* came forth. This *Butterfly* is very common in most Woods, but its *Caterpillar* is very rarely found.



L. III. ch. 1.

Cl. 3. S. a. 9.

Common Heath in Flower.

Erica Vulgaris. Park.

L. 3. CH. 1.

X.

CL. 2. S. a. 10.

THE WILLOW-BUTTERFLY. About the Middle of *August*, 1748, two of this Species of *Butterfly* were taken near *Camberwell*, in *Surry*: But in all my Practice I have never seen any of them in the Fields; so that they were look'd upon as very great Rarities. They are very common in *Germany*, and Mr. *Rosel* tells us, the *Caterpillar* feeds on *Willow*, and may be found all the Summer. The *Caterpillar* and *Chrysalis*, in the Plate, are taken from a Draught by Mr. *Rosel*.

See *Rosel*, Cl. 1, Tab. 1.



L. III. ch. I.

Cl. 3. S. a. 10.

The White Dog-Rose.

Rosa arvensis candida. C. B.

L. 3. CH. 1.

XI.

CL. 3. S. a. 11.

THE SMALL FRITILLARY-BUTTERFLY. About the 10th of April, 1741, I took upon the Ground upwards of one Hundred of these Caterpillars, in Cain Wood. I gave them to eat a Variety of Growths (which

I gathered on the Spot where I found the *Caterpillars*) but they were so restless and uneasy under Confinement, that they seemed in continual Motion, neither would they eat any of the Food I gave them. On the 18th of *April* ten of the *Caterpillars* fastened themselves up by the Tail, in order to their changing into the *Chrysalis* State, (the rest being dead or gone away) and on the 3d of *May* following the *Flies* were bred. This *Butterfly* is to be taken in Woods and Grass-Fields adjacent to Woods, in the Month of *May*.



L. III. ch. I.

Cl. 3. S. a. 11.

The Variegated Ketmia.

Ketmia Syrorum Flore variegato. J. R. H.

THE GREAT FRITILLARY-BUTTERFLY, with SILVER SPOTS. I take this *Fly* to be of the Class under Consideration. On the 15th of *July*, 1748, I had three Eggs laid by such a *Fly* as is represented in the Plate at Fig. 1, and on the 5th of *August* the young *Caterpillars* came forth; which, being examined with a Microscope, appeared to be of the Size represented. They were of a Flesh Colour, with Rows of Black Spots on each Joint like the *Caterpillars* of the *Emperor Moth*, and out of each Spot grew Hairs of a sandy Colour. The Eggs were beautifully fluted down the Sides, were flat at the Bottom, and had a glutenous Moisture upon them, which occasioned their sticking fast wherever the *Fly* chose to leave them. The *Caterpillars* on this present 10th of *February*, 1748-9, seem to be alive, but are very small, and, I believe, have eat nothing all the Winter, though they have had Grass given them, which I take to be their proper Food. The *Fly* frequents Woods and Meads, and loves to settle on *Blackberry Blossoms*, *Thistles*, &c. At the Beginning of *July* they may be taken with your Net; but as no *Butterfly* is more swift in Flight, you must attend till they settle, then be nimble and you may catch them.



L. III. ch. I.

Cl. 3. S. a. 12.

The Scorpion Sena.

Emerus Cæsalpin.



C H A P. II.

C A T E R P I L L A R S shaped like Wood-Lice.

C L A S S I.

S E C T. *a*.

L. 3. CH. 2.

I.

CL. 1. S. *a*. 1.

THE PURPLE HAIR-STREAK-BUTTERFLY. The *Caterpillar* feeds on *Oak-Leaves*, becomes a *Chrysalis* at the Beginning of *June*, and in a Month after the *Fly* appears. This *Butterfly* is not so easily taken as some others, for when it is not feeding it usually settles high. I have taken the greatest Numbers early in a Morning, when I have found them feeding on the Blossoms of the *Blackberry*.

See *Albin*, P. 57. *Rosel*, Cl. 2. Tab. 9.



L. III. ch. II.

Cl. 1. S. a. 1.

The Oak-Tree.

Quercus.

L. 3. CH. 2.

II.

CL. 1. S. a. 2.

THE BROWN HAIR-STREAK-BUTTERFLY. The *Caterpillar* of this *Butterfly* is seldom found: which, I believe, is owing to the oddness of its Shape and Colour, which are exactly represented in the

Plate. I took four of them by beating the *Black-Thorn*, and fed them on the same till the Middle of *June*, at which Time they changed into the *Chrysalis*, and the *Fly* was bred in the Middle of *August*. This *Butterfly* delights to settle on the *Maple-Tree*, &c. by beating the Branches whereof the *Fly* will rise, and may be taken in your Net.



L. III. ch. II.

Cl. 1. S. a. 2.

The Black-Thorn.

Prunus Sylvestris. Germ. Emac.

THE GREEN BUTTERFLY . Mr. *Dandridge* took the *Caterpillar* of this *Fly* at the Beginning of *July*, feeding on the inside of the *Blackberry* Buds, in which it made a small Hole, whereby it entered, and eating all the inward Part, left the Buds seemingly entire: It fed on them till the Middle of *July*, and then changed to a *Chrysalis*; and in the Middle of *April*, the Year following, the *Fly* was bred. This *Butterfly* is most common in and near Woods; it delights to settle on Boughs four or five Feet from the Ground, by moving which the *Fly* will rise, and if not much frightened will settle on or near the same Place again, by which Means you may take it with your Net. The best Time to catch this *Butterfly* is from the first to the fifteenth of *May*.



L. III. ch. II.

Cl. 1. S. a. 3.

The Blackberry, with its Fruit and Blossom.

Rubus, major, fructu nigro. J. B.



C H A P. III.

Of BUTTERFLIES whose Generation is not yet known.

C L A S S I.

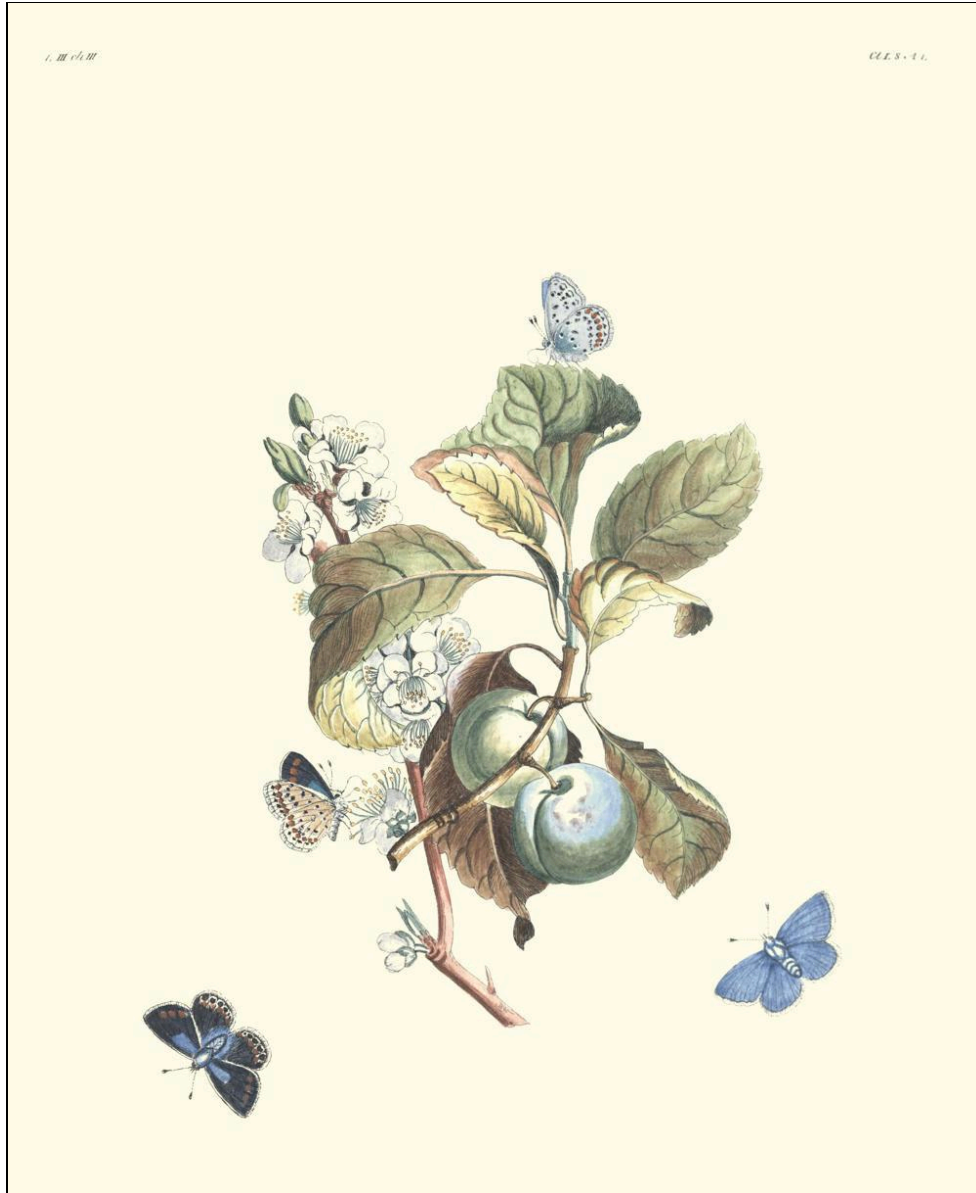
S E C T. A.

L. 3. CH. 3.

I.

CL. 1. S. A. 1.

THE BLUE ARGUS-BUTTERFLY breeds twice a Year: The first Brood appears in *May*, the second at the End of *July* and Beginning of *August*. This *Fly* is very common in most *Hay-Fields* at those Times, and is easily taken. As for the *Caterpillar* and *Chrysalis*, I have never yet heard that they have been discovered by any Body.



L. III. ch. III.

CL. 1. S. A. 1.

The Green Gage-Plumb.

Prunus fructu rotundo e viridi flavescente, carne dura suavissima. Miller.

L. 3. CH. 3.

II.

CL. 1. S. A. 2.

THE PURPLE HIGH-FLYER, or EMPEROR OF THE WOODS.
Neither the *Caterpillar* nor *Chrysalis* of this charming *Fly* has been yet

discovered, although sought after with the utmost Diligence for several Years past. The *Butterfly* appears at the End of *June* and Beginning of *July*, and may be taken in *Comb-Wood* in *Surry*, about *Westram* in *Kent*, and in other Places. It flies like a Hawk, delighting to soar aloft and skim in the Air. When it settles it is usually on some extreme Part of the *Oak*, *Hasle*, or *Ash-Tree*: and what is very singular, I myself have seen twenty of them taken on the same Branch one after another, for although the *Fly* seems to be extremely wild whilst on the Wing, yet, when settled, you may lay your Net over it with little Trouble.



L. III. ch. III.

Cl. 1. S. A. 2.

The Oak *with its* Acorns.

Quercus.

F I N I S.

I N D E X to the *F L I E S* .

A.

Admirable Butterfly. [55](#)

Angle Shades. [3](#)

Argus Butterfly.

———— great. [53](#)

———— Wood. *ibid.*

———— blue. [63](#)

Argus Moth.

———— Buff. [37](#)

Arrach Moth.

———— July. [36](#)

———— Wild. [3](#)

B.

Bindweed Hawk. See Hawk Moth.

Black Arches. [19](#)

Black-thorn Moth. [6](#)

Black Tussock. See Tussock-Moth.

Bramble Moth. [26](#)

Brimstone Butterfly. [48](#)

Brimstone Moth. [39](#)

Brindled Beauty. [35](#)

Brindle Moth. [43](#)

Broom Moth. [4](#)

———— green. [37](#)

Buff-Argus. See Argus Moth.

Buff Moth.

———— spotted. [20](#)

Buff Tip. [21](#)

Burnet Moth. [46](#)

C.

Cinnabar Moth. [26](#)

Codling Moth. [5](#)

Comma Moth. [57](#)

Cleifden Nonpareil. [45](#)

Cream-spot Tyger. See Tyger Moth.

Crimson Underwing. See Underwing Moth.

Curran Moth. See large Magpye.

D.

Dagger Moth. [28](#)

December Moth. [23](#)

Drinker. [27](#)

E.

Ealing's Glory. [12](#)

Egger.

—— Grass. [23](#)

—— great. [22](#)

—— small. [25](#)

—— small Oak. [44](#)

Elephant, or, Ladies Bedstraw Moth. [11](#)

———— small. [7](#)

Emperor Moth. [15](#)

Emperor of the Woods, or, Purple High-Flyer. [63](#)

Ermine Moth.

——— great. [20](#)

——— small. [4](#)

Eyed Willow Hawk. See Hawk Moth.

F.

Fox-coloured Moth. [25](#)

Fritillary Butterfly.

——— great. [57](#)

——— ——— with silver Spots. [59](#)

——— Heath. [58](#)

——— Plantain. *ibid.*

——— small. [59](#)

G.

Gipsy Moth. [20](#)

Glory of Kent. [45](#)

Goat Moth. [15](#)

Gold-spot Moth. [8](#)

Gooseberry Moth. [42](#)

Green Broom Moth. See Broom Moth.

Green Butterfly. [62](#)

Green Moth with silver Lines.

H.

Hair-streak Butterfly.

——— Purple. [61](#)

——— Brown. *ibid.*

Hawk Moth.

——— Bindweed, or, Unicorn Moth. [9](#)

———— Eyed Willow. [10](#)
———— Jessamine. [9](#)
———— Lime; or, Olive Shades. [10](#)
———— Poplar. [11](#)
———— Privet. [10](#)

Hawthorn Moth. [40](#)

J.

Jessamine Hawk. See Hawk Moth.

July Arrach Moth. See Arrach Moth.

July Sallow Moth. See Sallow Moth.

L.

Lackey Moth. [21](#)

Lady of the Woods; or, Orange Tip. [51](#)

Ladies Bedstraw Moth. See Elephant.

Lappit Moth. [27](#)

———— Wild Pine Tree. [29](#)

Lime Hawk. See Hawk Moth.

Lime Moss Moth. [37](#)

M.

Large Magpye, or Curran Moth. [42](#)

Marmoris, or marble Butterfly. [52](#)

Meadow Brown. *ibid.*

Mother of Pearl Moth. [24](#)

Mottled Umber. [36](#)

N.

Nut-tree Tussock. See Tussock Moth.

O.

Oak-Moth.

——— yellow July. [24](#)

——— small Green. [4](#)

——— scallop-wing'd. [6](#)

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October Moth. [38](#)

Olive Shades. See Lime Hawk.

Orange Moth. [40](#)

Orange Tussock. See Tussock.

Orange Tip. See Lady of the Woods.

Ozier, or Pebble Moth. [12](#)

P.

Painted Lady. [56](#)

Peacock Butterfly. [55](#)

Pease Blossom. [3](#)

Pebble Moth. See Ozier Moth.

Plumb-Tree Moth. [17](#)

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Privet Hawk. See Hawk Moth.

Province Rose Moth. [5](#)

Purple High-Flier. See Emperor of the Woods.

Puss Moth. [13](#)

R.

Ranunculus Moth. [7](#)

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Richmond Beauty. [39](#)

S.

Sallow Moth. [5](#)

———— July. [36](#)

Scallop-wing'd Moth. [39](#)

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Silver Y Moth. [34](#)

Spotted Buff. See Buff Moth.

Spotted Elm Moth. [38](#)

Spotted red and white Underwing. See Underwing.

Swallow-Tail Butterfly. [47](#)

Swallow-Tail Moth. [38](#)

Sword-Grass Moth. [8](#)

Sycamore Tussock. See Tussock Moth.

T.

Great Tortoise-shell Butterfly. [56](#)

Tussock Moth.

———— black. [30](#)

———— Nut-tree. [31](#)

———— Orange. *ibid.*

———— Red-spot. [30](#)

———— Sycamore. [32](#)

———— yellow. [30](#)

Tyger Moth.

———— Cream-spot. [18](#)

———— great. *ibid.*

———— Scarlet. [19](#)

———— Wood. [24](#)

U.

Underwing Moth.

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- great yellow. [2](#)
- spotted red and white. [23](#)
- Willow red. [17](#)

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Water Betony Moth. [7](#)

White Butterfly with black Veins. [49](#)

————— with green Veins. [50](#)

White Garden Butterfly.

————— large. [49](#)

————— small. [50](#)

White Satin Moth. [21](#)

Wild Arrach Moth. [3](#)

Wild Rose Moth. [14](#)

Willow Butterfly. [58](#)

Willow red Underwing. See Underwing Moth.

Wood Tyger. See Tyger Moth.

Y.

Yellow Tail. [28](#)

Great yellow Underwing. See Underwing Moth.



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Almond Tree. *Amygdalus sativa*. [25](#)
————— double flowering. [50](#)

Apple Tree. *Malus*. [19](#)

Apricot. *Malus Armeniaca*.

———— Brussels. [6](#)

———— Masculine. [31](#)

———— Orange. [26](#)

Red Archangel. *Lamium rubrum*. [30](#)

Wild Arrach. *Atriplex*. [3](#)

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B.

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Bistort. *Bistorta*. [28](#)

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—— Common. [57](#)

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—— Witch. [38](#)

G.

Gooseberry Tree. *Grossularia*. [42](#)

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—— Bent. *Gramen Segetum*. [52](#)

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- great Fox-tail. *Gramen Alopecuroides*. [52](#)
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—— Variegated. *Ketmia Syrorum*. [59](#)

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